

FA9101-26-R-B011

Attachment 1

STATEMENT OF WORK (SOW)

**FACILITIES ACQUISITIONS FOR
RESTORATION AND
MODERNIZATION (FARM) III**

19 February 2026

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SECTION 1

DESCRIPTION OF SERVICES

1 INTRODUCTION

The Air Force Test Center's (AFTC) Arnold Engineering Development Complex (AEDC) is a national aerospace ground test facility that conducts tests, engineering analyses, and technical evaluations for research, system development, and operational programs of the Air Force (AF) and Department of Defense (DoD), other Government agencies, and industry. Using ground test facilities and computational engineering, AEDC supports propulsion, aerodynamic, reentry, trans-atmospheric, and space-flight systems testing. This testing underpins the technical knowledge required for the development and qualification of key warfighter aerospace weapons. Testing is performed in an environment that simulates operational conditions. AEDC also performs research to develop new technology for advanced test facilities, test techniques, and measurement methodologies associated with ground tests.

This Statement of Work (SOW) sets forth basic requirements for construction and engineering activities supporting the Facilities Acquisitions for Restoration and Modernization (FARM) III. Performance on this contract will be primarily required at Arnold Air Force Base (AFB) TN. Performance will also be required at additional AEDC Geographically Separated Units (GSUs), including but not limited to: National Full Scale Aerodynamics Complex (NFAC), Mountain View CA; Tunnel 9, White Oak MD; and the 704th Test Group at Holloman AFB NM; to include the 704th's GSUs, OL-AA located at Kirtland AFB NM, OL-AC located at Wright Patterson AFB OH, and the 586th Flight Test Squadron at White Sands Missile Range NM. Many of the provisions described in this SOW are specific to Arnold AFB TN where most of the work will be performed. Additional provisions may be included and/or modified on task orders (TOs) for AEDC GSUs.

This SOW is organized and developed to provide maximum flexibility in the use of this contract vehicle. Basic requirements described herein are further refined through the use of individual project TOs. TOs will clearly define the project requirements in terms of performance, appearance, criteria, and/or specification with sufficient detail that permits a contractor to provide a proposal with reasonable performance, cost and schedule risk. TOs will be issued for projects at various stages of design (e.g., no design, partial design, or full design) and construction.

The contractor shall function as an integral team member in support of the AEDC mission, to include sharing information with other AEDC contractors and cooperation with communities and other Government entities.

1.1 BACKGROUND AND OBJECTIVES

The FARM III effort acquires a variety of project design and construction services. The required capabilities are for specialized and unique mechanical; electrical; and instrumentation, data, and control (ID&C) systems required to restore and modernize AEDC's plant and infrastructure systems while minimizing the facility downtime and impact on existing test and development programs.

AEDC provides the most comprehensive set of aerospace ground test facilities in the world. Many of the individual test facilities are unique in the country or in the world. AEDC capabilities include very large- to medium-sized wind tunnels which cover the entire flight envelope from subsonic to hypersonic speeds. Simulated altitude testing for very large commercial-type turbofan engines and the world's most powerful fighter engines, as well as large rocket motors, can be conducted at AEDC. Facilities at AEDC can further simulate space environments and space vehicle reentry speeds and temperatures. AEDC currently operates at the following locations: Arnold AFB TN, Mountain View CA, White Oak MD, Edwards AFB CA, Vandenberg Space Force Base (SFB) CA, Hill AFB UT, Peterson SFB CO, and the 704th Test Group at Holloman AFB NM, to include the 704th's GSUs: OL-AA located at Kirtland AFB NM, OL-AC located at Wright Patterson AFB OH, and the 586th Flight Test Squadron at White Sands Missile Range NM.

The test facilities and supporting plant and infrastructure systems at Arnold AFB and its GSUs were primarily built in the 1950s and 1960s with a few additions in the 1990s. For decades, the restoration and modernization of test plant and test infrastructure was underfunded, resulting in diminished test capability and increased energy and maintenance costs. With the successful articulation of potential impacts to future system development, AF leadership secured funding for major investment beginning in fiscal year (FY)17 to begin to mitigate the long-term risk in support of next generation aerospace system development. The FARM I contract established a contract vehicle to provide execution of these investments until FY22; FARM II contract followed but will expire in FY27. With continued investment on the horizon, AEDC will require a contract vehicle (FARM III) to execute these requirements.

The Government will serve as the lead Program Manager (PM) with support from other AEDC support contractors for TOs executed under this contract. The contractor shall serve as the prime or lead for projects integrating multiple capabilities.

1.2 SCOPE OF WORK

The contractor shall provide all labor, materials, equipment, transportation, supervision, quality control, and other items necessary to design, manage and accomplish the work. The effort will be performed primarily at Arnold AFB TN, but will also be required at the GSUs referenced in paragraph 1.1 of this SOW.

The types of projects within this SOW generally include the Design/Build, Construction, and Two-Phase Design/Build of mechanical, electrical, and ID&C for all AEDC locations. Major capabilities, including design, to accomplish typical projects within this effort include, but are not limited to:

Mechanical

- M1 Air duct, expansion joints and control/process valves ($\geq 36"$ Ø and 150 psig)
- M2 High pressure duct/piping and valves (typical 4" to 8"Ø, ≥ 4000 psig)
- M3 Large pressure vessels ($\geq 2,000$ psig; $\geq 10K$ gallons)
- M4 Desiccant bed air dryers (≤ 300 psia and ≥ 200 lb/s)
- M5 Turning vanes for large air ducting (up to 50-ft Ø; 0.5-160 psia)

- M6 Cooling water/raw water pumps (300-50,000 gpm; 30-4,000 hp), filters/strainers, valves (control/process) and distribution lines (typ. 6" to 90" Ø)
- M7 Hydraulic systems with high flow and pressure requirements (20-200 gpm [typ. 2 pumps @ 100 gpm ea] and 2,500 psig)
- M8 Large steam distribution systems (60kpph boilers, and piping <14" Ø)
- M9 Chillers and refrigeration equipment (300-5,000 refrigeration tons)
- M10 Large heat exchangers: air to air; water to air; shell and tube, etc. (ΔT up to 325 °F)

Electrical

- E1 Interior and exterior lighting for industrial buildings
- E2 Medium voltage gas insulated switchgear ($\leq 13.8/6.9/2.4$ kV)
- E3 Unit substations (typical 13.8/6.9/2.4kV)
- E4 Motor Control Centers ($\leq 600V$)
- E5 Transformers ($\leq 161/13.8/6.9kV$)
- E6 Uninterruptable Power Supply (UPS)
- E7 Ducting/vaults/ cables (161, 13.8 and 6.9 kV)
- E8 Motor exciters
- E9 Oil or gas circuit breakers (typical 1,200/2,000/3,000 amp and 550/750kV basic insulation level [BIL])
- E10 Power panels/conduit/wiring ($\leq 480V$)
- E11 Variable frequency drives for motors ($\geq 600V$ and $\leq 69kV$)
- E12 Facility Wide Lightning Protection Systems

ID&C

- I1 Programmable Logic Controller (PLC)-based industrial control systems
- I2 High speed ($< 10ms$ Loop Time) facility control systems
- I3 Controls software development and modification
- I4 Secure control systems
- I5 Data logging system design and integration
- I6 Control systems instrumentation design and integration

1.3 TO TYPES

The FARM III effort will be competed as a Multiple Award Construction Contract. This Indefinite-Delivery Indefinite-Quantity (IDIQ) contract will be for the execution of a broad range of design/build, two-phase design/build, and construction projects. The performance of work will be accomplished by the contractor through individual TOs. The project scope on each TO will be identified by the Government. TOs will be competed by Fair Opportunity Proposal Request (FOPR) using either a Lowest Price Technically Acceptable (LPTA) or a full tradeoff source selection process. The process to be utilized will be defined in each TO. TOs will be of one of the following types:

- a) Design/Build: The contractor will be provided with a TO. The contractor shall initiate and complete the design/build project execution in accordance with (IAW) the TO requirements. The level of design required may be tailored by the Government depending on the TO complexity.

b) Construction: The contractor will be provided a complete design document package to execute the construction IAW TO requirements. The package will include the requirements document, drawings, technical specifications, and any other preliminary data required to complete the TO.

c) Two-Phase Design/Build: The contractor will be provided with a TO. Multiple contractors will be selected to complete a percentage of the design IAW the TO requirements. The Government will down-select to one contractor to complete the design and build portion of the design/build TO. The level of design required may be tailored by the Government depending on TO complexity.

1.4 PROJECT ADMINISTRATION

The contractor shall perform management, planning, and reporting to effectively secure and monitor prime and subcontractor work.

1.4.1 Contractor Organization: If self-performing capabilities are not practicable, the prime contractor shall form a teaming arrangement with a mechanical, electrical, or ID&C construction or engineering design firm to meet the full range of requirements of this contract. The contractor shall provide comprehensive design capabilities including assessment, analysis, planning, management, and quality control to ensure a completely functional and finished product.

1.4.1.1 Project Management: The contractor shall be responsible for the overall management of each TO performed under this contract. The contractor shall be responsible for timely status reporting as specified in the TO. The contractor shall ensure that quality work is completed on schedule and within the allocated budget. The contractor shall ensure that, during construction, project leadership is on-site no less than monthly to monitor TO progress.

1.4.1.2 Project Supervision: The contractor shall identify the primary point of contact (POC) during all phases of the TO. The contractor shall notify the CO in writing of the intended project superintendent before the issuance of the Notice to Proceed (NTP). If the superintendent is changed during the TO Period of Performance (PoP), the contractor shall notify the CO in writing of the intended replacement. The designated superintendent or delegated representative shall be on-site anytime construction work (including work executed by a subcontractor) is being performed. The contractor shall provide the Contracting Officer (CO) and the Contracting Officer Representative (COR) with the names of a primary and alternate POC's telephone numbers in case of emergency. Requests for emergency on-site presence shall be met within 24 hours.

1.4.2 Removal of Contractor's Employees: The contractor agrees to utilize only experienced, responsible, and capable people in the performance of work. The CO may require that the contractor remove employees who endanger persons or property, whose continued employment under this contract is inconsistent with the installation Commander's safety and security interests of good order and discipline, or who are determined by the CO or COR as being incapable of rendering the services IAW this contract.

1.4.3 Business Relations: The contractor shall successfully integrate and coordinate all activities needed to execute the TO. The contractor shall manage the timeliness, completeness, and quality of problem identification. The contractor shall provide corrective action plans,

proposal submittals, timely identification of issues, and effective management of subcontractors. The contractor shall coordinate and cooperate closely with associate contractors involved in the execution of AEDC's mission.

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SECTION 2

2 GENERAL INFORMATION

2.1 GENERAL TERMS

This contract is a firm-fixed-price (FFP), IDIQ type and will include a wide variety of individual specific elements of Construction, Design/Build, and/or Two-Phase Design/Build. During the contract period, the Government will identify Construction, Design/Build, and/or Two-Phase Design/Build tasks required to complete specific projects, and the CO will negotiate and issue individual TOs to the contractor to perform the projects. The contractor shall furnish all materials, equipment, and personnel necessary to design, manage, and accomplish each TO as required. The Government will issue an NTP for each TO placed against the basic contract.

2.2 QUALITY CONTROL PLAN (QCP)

For each TO, the contractor's QCP shall be developed and submitted to the CO for approval. The QCP shall be submitted within 14 calendar days after contract award. The QCP shall cover both design and construction activities. The contractor shall designate, in writing, on-site personnel to document and resolve all quality control problems and accomplish daily quality control inspections. IAW the contractor's Quality Management System, the contractor's QCP shall establish a method to identify poor performance and specify appropriate corrective action as it relates to the specific TO. The QCP shall become a compliance document. Project personnel shall interface with Government personnel to ensure that the QCP functions properly.

2.3 PROJECT SCHEDULE

The contractor shall provide a project schedule for each TO. The project schedule shall be developed as time-phased Gantt charts using Microsoft Project or equivalent, unless otherwise specified in the TO. The project schedule shall depict activities, milestones, outages, start and completion dates, predecessor and successor relationships, and the critical path. Updates to the project schedule shall be made as required throughout the project PoP. Schedule updates showing activity extending beyond the PoP are not an extension of the PoP. PoP extensions will only be granted through TO modifications issued by the CO. Schedule reporting requirements will be identified for each TO; typically TOs will require an updated schedule to be delivered in its native format on or before the 15th of every month.

The contractor shall maintain individual TO schedules and a consolidated schedule for all active TOs it is responsible for. The contractor's consolidated schedule may be requested by the Government to assist in coordinating operations and TO work as necessary.

2.4 INTEGRATED SCHEDULING

The Integrated Scheduling process is used to schedule test, maintenance and capital improvement projects conducted at AEDC. The contractor shall coordinate outages with the Government Representative to be incorporated into AEDC's Integrated Scheduling process to plan and execute work.

An outage represents a specified period of time that an asset or assets will be unavailable for operations. The outage process gives visibility to activities that impact AEDC operations.

The FARM III TOs will occur in a defined window of AEDC's highly active test schedule. Outage requests are used to coordinate work which can affect execution of testing. Outages when required by the contractor for the test facilities, plant equipment or support utilities/areas, are expected to be minimized as well as coordinated in advance with the Mission Integration Group (MIG) through the Government Representative. Any required verification testing shall also be identified at that time and be included within the outage request. Replacement/refurbishment of existing equipment and installation of new systems must occur in a manner and fashion to provide maximum continuity of operations within the AEDC MIG Integrated Schedule.

The contractor shall coordinate schedule requests with the Government Representative to accomplish any verification testing (operational checkouts) required by the TO. The Government Representative will coordinate with the MIG for operational checkouts of major pieces of equipment which impact testing or require operator support. These requirements must be identified as early as possible in order to facilitate efficient scheduling.

All outage requests, including but not limited to, utility interruptions and road work, shall be planned in order to accomplish the work as defined in the TO. The outage requests shall be submitted in writing by the contractor to the Government Representative. The Government Representative will coordinate with the MIG to schedule the outage. Scheduling alternatives will be identified by the MIG and provided to the Government Representative to coordinate with the contractor. For purposes of scheduling, AEDC maintains a 24/7 testing schedule. The contractor shall perform preliminary work to the greatest extent possible before beginning the outage and shall determine in advance that the system can be placed back in operation prior to expiration of the outage period. The contractor shall provide the advanced notice for outages to the Government Representative as follows;

- a) For outages of 3 days or less, the outage shall be requested no later than (NLT) 60 days in advance.
- b) For outages greater than 3 days and up to 7 days, the outages shall be requested NLT 120 days in advance.
- c) For outages greater than 7 days, the outages shall be requested NLT 180 days in advance.

The contractor's outage request shall include the following:

- 1) Type of outage;
- 2) Access or service to be disrupted;
- 3) Areas and/or facilities affected;
- 4) Expected duration of outage;
- 5) Preferred outage date with two alternate dates;
- 6) POC and telephone number; and

7) Lists of materials and equipment to be used.

The Government will make every effort to schedule the outages during the contractor's preferred dates; however, mission testing may preclude the preferred dates. The Government will provide approval for the outage to the contractor a minimum of 14 days before the actual outage. Prior to this 14 day lead time, the Government reserves the right to move outage windows based on mission requirements. In addition, short duration outages may be requested at any time and approved by the Government Representative, subject to the AEDC test schedule and coordination with Integrated Scheduler. The requirements for specific types of outages are as follows:

- a) Pile driving: Outages may be required depending on the location and proximity to critical AEDC systems. During performance of the contract, the Government Representative may provide additional times when outages may be required because of testing in adjacent base-operated test cells.
- b) Base Utilities: Outages will be granted only on weekends between 0500 Saturday and 0700 Monday.
- c) Roads: The contractor shall provide minimum notice of 14 days prior to closing any road. Roads shall not be closed at more than one point at the same time.

For outages seven days and longer, the contractor shall not exceed the number and length of outages specified in the TO.

2.5 MEETINGS

The contractor shall attend and/or support meetings and teleconferences to discuss technical requirements, issues and project progress, and status. Meeting requirements will be identified with each TO. The contractor shall prepare and submit for review, an agenda, presentation materials, and minutes for meetings hosted by the contractor, as applicable. Meeting minutes shall include any action items and identify the responsible person for action items.

2.6 CONFIGURATION MANAGEMENT

The contractor shall document and turnover all newly developed, revised, or re-engineered software, drawings, and documentation to the Government as indicated in the individual TO and in a format specified by the Government. The contractor shall utilize configuration management throughout the development and deployment of software, drawings, and documentation.

2.7 TEST, MEASUREMENT AND DIAGNOSTIC EQUIPMENT (TMDE)

The contractor shall document any TMDE provided by the contractor as part of the TO and coordinate with the Government, as indicated in the individual TO.

2.8 MASTER WORK PERMITS (MWPs)

The contractor shall be responsible for coordinating a Government-issued MWP prior to performing any site work, especially digging of any type. The Government will assist the contractor in the coordination of the first MWP request and provide details as to how coordination is to be accomplished. Thereafter, Government assistance will be limited to an as-needed basis in the event of unusual circumstances. It will be the contractor's responsibility to

coordinate completion of the MWP and request to have existing utilities located and marked as indicated on the completed form, prior to the beginning of digging operations in the individual areas. It will also be the contractor's responsibility to maintain the markings and handle any required extensions of MWP requests. This coordination is anticipated to take approximately 10 working days to complete per request.

2.9 SUBMITTAL PROCEDURE

This section specifies procedural requirements for non-administrative submittals, including but not limited to shop drawings, product data, manufacturer's certificate, design data, calculations and verifications, manufacturer's instructions, manufacturer's field service reports, samples, O&M manuals, and other miscellaneous work-related submittals. These submittals are required to amplify, expand, and coordinate other information contained in the contract.

The CO may request submittals in addition to those specified when deemed necessary to adequately describe the work covered in the respective sections of the specification requirements document.

2.9.1 Listing: Each TO will summarize items required for submittal. The list of submittals is included for the convenience of users of the TO documents. The listing may not be all inclusive and additional item submittals may be required. The contractor shall submit a copy of the summarized submittal listing with calendar dates assigned to each item submittal indicating when the submittal will be received by the CO. The submittal listing shall be submitted within 14 calendar days after the design is approved for design/build TOs or 14 calendar days after receiving award for construction TOs.

2.9.2 Risk: The contractor shall not proceed with any part of the work covered by an item submittal including purchasing, fabricating, and delivering until approval is received. Any machinery, equipment, material, or articles purchased, fabricated, or otherwise used that do not have the required advance approval that are installed or used are at risk of subsequent rejection by the Government. Any fabrication or other work performed in advance of the receipt of accepted item submittals and approvals shall be entirely at the contractor's risk and expense.

2.9.3 Transmittal Timing: The contractor shall coordinate the preparation and processing of item submittals with the performance of the work. Unless otherwise stated in the TO requirements document, the contractor shall prepare and transmit each item submittal to the CO a minimum of 21 calendar days prior to the performance of related work and other applicable activities. The contractor shall transmit related item submittals for the same unit of work so that processing will not be delayed by the Government's need to review submittals concurrently for coordination. No delay damages or time extensions will be allowed for time lost in late submittals.

2.9.4 Content: Each submittal shall be complete and in sufficient detail to allow ready determination of compliance with contract requirements. All submittals, which are generic and list more information than is specifically required, shall be marked to identify required information. The contractor shall complete an AF Form 3000 and attach it as a cover sheet for each submittal. The contractor may include multiple item numbers on an AF Form 3000.

2.9.5 Documentation: All typed material such as specifications, reports and studies shall be in Microsoft Office format (e.g., .doc, .docx, .xls, .xlsx, .ppt, .pptx).

2.9.6 Language: All item submittals shall be written in English, including documents, notes on drawings, sketches, and/or samples, calculations, manuals, and all other instances of written text and communication.

2.9.7 Units: All item submittals shall be marked or show dimensions and values in consistent United States (US) Customary units.

2.9.8 Review Time: The contractor shall allow sufficient time so that TO performance will not be delayed as a result of the time required to properly process submittals, including time for resubmittals, if necessary. The contractor shall allow 21 calendar days for initial Government processing of each submittal. No extension of time will be authorized because of the contractor's failure to transmit submittals to the Government sufficiently in advance of the work.

2.9.9 Contractor Certification: All submittals shall be carefully reviewed by an authorized representative of the contractor prior to submission to the Government. Each submittal shall be dated, signed, and certified by the contractor as being correct and in strict conformance with the TO documents. No consideration for review by the Government of any contractor's submittal will be made for any items which have not been certified by the contractor. All noncertified submittals will be returned to the contractor without action taken by the Government and any delays caused thereby shall be the responsibility of the contractor.

2.9.10 Deviations: Should any item submittals required by the TO show deviations from the TO requirement, the contractor shall make specific mention of such deviations in the letter of transmittal. The letter shall state cost effects and product and system limitations which may adversely affect the work, in order that, if acceptable, suitable action may be taken for proper adjustment of the contract. Otherwise, the contractor will not be relieved of the responsibility for executing the work IAW the TO. The contractor shall clearly mark the proposed variation in all documentation and specifically point out deviations from the TO requirements in transmittal letters. Failure to point out deviations may result in the Government requiring rejection and removal of such work at no additional cost to the Government. Deviations from TO requirements require Government approval and will only be considered when advantageous to the Government. When proposing deviations, the contractor shall deliver a written request to the CO, with documentation of the nature and features of the deviation and why the deviation is desirable and beneficial to the Government. If lower cost is a benefit, the contractor shall include an estimate of the cost savings. In addition to documentation required for the deviation, the contractor shall include the submittal information required for the item. The contractor shall allow an additional 14 calendar days beyond normal submittal review period for consideration by the Government of submittals with deviations.

2.9.11 Approved Submittals: The part of the work covered by the approved item submittal may proceed provided it complies with the requirements of the TO documents. Final acceptance will depend upon that compliance. The term "Approved" shall only indicate that there is no exception taken to the submittal. Approval of the item submittal shall not be construed as a complete check and indicates only that the general method or other information appears to meet the TO requirements. Approval does not relieve the contractor of the responsibility for any error

which may exist. After item submittals have been approved by the CO, no resubmittal for the purpose of substituting materials or equipment, or changes to any other information, will be considered unless accompanied by an explanation of why a substitution or change is necessary.

2.9.12 Disapproved Submittals: The contractor shall correct disapproved item submittals and resubmit for approval within timeframe noted. If no date is given, the contractor shall submit the corrected submittal within 14 calendar days. If the contractor considers any correction or notation on the returned submittal to constitute a change to the TO drawings, specification, or requirements document, the contractor shall notify the CO.

2.9.13 Responsibility: The Government's review of contractor submittals will not relieve the contractor of the entire responsibility for the accuracy of details and dimensions and conformance to the specifications. The contractor shall assume all responsibility and risk for any mistakes and/or costs due to any errors in submittals.

2.9.14 Inconsistencies: If a conflict or inconsistency arises between an approved item submittal and the TO documents, the TO documents shall govern.

2.10 SPECIFIC SUBMITTAL DESCRIPTIONS AND REQUIREMENTS

Submittal requirements for individual units of work are specified in the TO and should comply with the following requirements for each type of submittal. The contractor shall electronically submit all documents in a file type and media format defined by the Government in the TO unless otherwise specified.

2.10.1 Shop Drawings: These are technical drawings and data prepared for each TO including fabrication and installation drawings, setting and seaming diagrams, and coordination drawings (for use on-site). Shop drawings include drawings, diagrams, and schedules specifically prepared to illustrate some portion of the work, diagrams, and instructions from a manufacturer or fabricator for use in producing the product. Shop drawings also include aids to the contractor for integrating the product or system into the project, or drawings prepared by or for the contractor to show how multiple systems and interdisciplinary work will be coordinated. Information required on shop drawings includes dimensions, identification of specific products, and materials which are included in the work, compliance with specified standards, and notations of coordination requirements with other work. The contractor shall provide special notations of dimensions that have been established by field measurements and shall highlight, encircle, or otherwise indicate deviations from the TO documents on the shop drawings.

2.10.2 Product Data: This data includes standard printed information on manufactured products, including manufacturers' product specifications illustrating size, physical appearance, and other characteristics of materials; installation instructions; standard color charts; catalog cuts; illustrations; schedules; standard wiring diagrams; standard product operating and maintenance manuals; and samples of warranty language when the TO requires extended product warranties. General information required specifically as product data includes manufacturers' standard printed recommendations for application and use, compliance with recognized standards of trade associations and testing agencies, the application of their labels and seals (if any), special notation of dimensions which have been verified by way of field measurement and special coordination requirements for interfacing the material, product, or system with other work.

2.10.3 Samples: These are physical examples of work including, but not limited to, swatches showing color, texture, and pattern; color-range sets; and units of work to be used for independent inspection and testing. Samples include fabricated or unfabricated physical examples of materials; equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged; color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project; and field samples and mock-ups constructed on the project site to establish standards by which the ensuring work can be judged. This includes assemblies or portions of assemblies, which are to be incorporated into the project and those which will be removed at conclusion of the work. The contractor shall submit samples for the CO's visual review of general kind, color, pattern, and texture for a final check of the coordination of these characteristics with other related elements of the work and for quality control comparison of these characteristics between the final sample submittal and the actual work as it is delivered and installed. The contractor shall submit one sample for each physical example or assembly.

2.10.4 Design Data: Design data includes design calculations, mix designs, analyses, or other data pertaining to a part of work. The contractor shall refer to individual sections of the TO for required quantities, formats, and signatures or certifications required.

2.10.5 Test Plans and Reports: Test plans shall include planned testing, including a description of the test, equipment and supplies needed, and step-by-step notation of test activities and tasks. Test reports include reports signed by an authorized official of a testing laboratory that a material, product, or system identical to the material, product, or system to be provided has been tested IAW the specified requirements. Testing shall have been within three years of date of contract award for the project. Reports also include findings of tests required to be performed by the contractor on an actual portion of the work or prototype prepared for the project before shipment to the job site, findings of tests made at the job site or on a sample taken from the job site, investigation reports, daily logs and checklists, and final acceptance test and operational test procedures.

2.10.6 Certificates: These include statements printed on the manufacturer's letterhead and signed by the authorized officials of the manufacturer of a product, system, or material attesting that the product, system, or material meets specification requirements. Certificates include documentation required of the contractor, or of a manufacturer, supplier, installer, or subcontractor through the contractor, to further demonstrate the quality of orderly progressions of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications. Examples include confined space entry permits and text of posted operating instructions. Certificates shall be dated after award of the project TO and clearly name the project.

2.10.7 Manufacturer's Instructions: The contractor shall provide preprinted material describing installation of a product, system, or material, including special notices and Material Safety Data sheets concerning impedances, hazards, and safety precautions.

2.10.8 Manufacturer's Field Reports: The contractor shall provide documentation of the testing and verification actions taken by a manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or

after installation, to confirm compliance with the manufacturer's standards or instructions. The documentation shall be signed by an authorized official of a testing laboratory or agency and shall state the test results, and indicate whether the material, product, or system has passed or failed the test.

2.10.9 O&M Data: The contractor shall provide data supplied by the manufacturer/system provider to the equipment operating and maintenance personnel, including manufacturer's help and product line documentation necessary to maintain and install the equipment, and needed by operations and maintenance (O&M) personnel for the safe and efficient operation, maintenance, and repair of the item.

2.10.10 Final Design Drawings and Documentation: The contractor shall provide final design documentation including drawings, specifications, estimates, and other required deliverables with no restrictions on the Government's use of the documentation. Electronic documentation submitted shall include every associated component file of drawings and other documentation, including raster images, jpeg, tiff, Excel, and any related items required to view, modify, or manipulate the electronic files. The contractor shall electronically provide all final design documentation and drawings in AutoCAD 2018 format or later and as Portable Document Format (PDF) files.

2.10.11 Record (As-Built) Drawings: The contractor shall submit record drawings showing final configuration of work accomplished. The contractor shall show all changes, additions, and deviations from the original TO drawings and documentation; including but not limited to, actual routing of field routing of cabling/conduit/piping, final Input/Output (I/O) counts, wiring details, and final connection details regardless of the level of design drawing detail. If no changes occur, certification shall be provided to that effect. Redlined hardcopies shall be maintained by the contractor during construction to accurately show as-built conditions during the progress of the job. Redline drawings are only acceptable as record (as-built) drawings when the contractor is not provided access to the original drawings in a format that is editable. If electronic, drawings shall be provided in AutoCAD 2018 format or later and as PDF files. The contractor shall additionally include any associated component file of the drawings, including raster images, jpeg, tiff, Excel, and any related items required to view, modify, or manipulate the electronic files. The contractor shall submit these drawings to the CO for approval prior to applying for final payment.

2.10.12 Miscellaneous Submittals: These are work-related, non-administrative submittals that do not fit in the previous categories, including the following:

2.10.12.1 Maintenance agreements: The contractor shall provide maintenance agreements electronically in a file and media format defined by the Government in the TO.

2.10.12.2 Survey data and reports: The contractor shall provide survey and data reports electronically in a file and media format defined by the Government in the TO.

2.10.12.3 Project photographs: The contractor shall provide project photographs electronically in a file and media format defined by the Government in the TO.

2.10.12.4 Keys and security devices: The contractor shall provide keys and other security protection devices as noted in the TO.

2.10.12.5 Tools, spare parts, and stock: The contractor shall provide maintenance tools, spare parts, and overrun or maintenance stock. Refer to individual TO specifications for required quantities of spare parts, extra and overrun stock, maintenance tools and devices, keys, and similar physical units to be submitted.

2.10.12.6 Warranties: The contractor shall provide warranties electronically in a file and media format defined by the Government in the TO.

2.11 REQUIRED INSURANCE

The contractor shall procure and maintain during the entire period of performance of each TO all required insurance. Additionally, the contractor shall submit to the CO all required insurance certifications within 10 days of the notice of award for construction work. At a minimum, the following insurance coverage must be maintained throughout TO performance:

<u>Type</u>	<u>Amount</u>
Workers' Compensation and Employers Liability Insurance	\$100,000
General Liability Insurance (Bodily Injury)	\$500,000 per occurrence
Automobile Liability	\$200,000 per person
	\$500,000 per occurrence (bodily)
	\$20,000 per occurrence (property)

SECTION 3

3 DESIGN

This design section describes the overarching design process, but the Government does not intend to require this full process for every TO. Individual TOs may be tailored based on the Government's assessment of the effort's cost, schedule, complexity, and risk factors.

3.1 SCOPE

3.1.1 This section outlines the required professional services to be provided by the contractor for a design/build TO under this contract. The contractor shall provide sufficient technical, supervisory, and administrative personnel to ensure the expeditious accomplishment of the design work specified as required for the duration of the contract.

3.1.2 Professional engineering (PE) services as defined by state laws in which the work is being performed will be required for design/build TOs. The primary professional disciplines include mechanical, electrical, and ID&C disciplines.

3.1.3 Other professional services of an engineering or architectural nature include, but are not limited to:

- a) Studies
- b) Investigations
- c) Surveying* and mapping
- d) Tests
- e) Evaluations
- f) Consultations
- g) Conceptual designs
- h) Value engineering
- i) Construction phase services
- j) Geotechnical reports and review
- k) Asbestos abatement, including work plans and specifications
- l) Lead based paint abatement, including work plans and specifications
- m) Preparation of operating and maintenance manuals

*Surveying services procured under this IDIQ shall be limited to services directly associated with the planning, design, or construction of real property improvements under TOs issued pursuant to this contract.

3.2 REFERENCES AND STANDARD DESIGN CRITERIA

Design and construction shall conform to latest editions and supplements of the publications, listed in Attachment 1, available at the award of the TO.

3.3 REQUIREMENTS

3.3.1 All work shall be technically accurate before submission.

3.3.2 The contractor is responsible for the quality, technical accuracy, and the coordination of all design, drawings, specifications, and other services supplied under this contract. The contractor shall, without additional compensation, correct or revise any errors, deficiencies, or inadequacies in its designs, drawings, specifications, and other services.

3.3.3 Neither the Government's review, approval, acceptance, nor payment for the services required under this contract, shall be construed to operate as a waiver of any rights under this contract or of any cause of action arising out of the performance of this contract. The contractor shall remain liable to the Government IAW applicable law for all damages to the Government caused by the contractor's negligent performance of any of the services rendered under this contract.

3.3.4 When Automated Data Processing Systems (ADPS) are used to perform design calculations, the design analysis shall include descriptions of the computer programs used and copies of the ADPS input data and output summaries. Preparation of the descriptions that must accompany each set of ADPS printouts shall include the following:

- a) Explain the design method, including assumptions, theories, and formulas.
- b) Include applicable diagrams, adequately identified.
- c) State exactly the computation performed by the computer.
- d) Provide all necessary explanations of the computer printout format, symbols and abbreviations.
- e) Use adequate and consistent notation.
- f) Provide sufficient information to permit manual checks of the results.

3.3.5 Engineers or architects registered to practice in the particular professional field shall be involved in preparing, reviewing, and approving the design. The engineer or architect for the contractor shall be an individual, firm, or any duly authorized representative thereof, professionally engaged in the practice of engineering and/or architecture, qualified by registration and experience and licensure in the applicable state to offer the services specified to the public.

3.3.6 The contractor shall perform a code analysis for each TO and submit a list of applicable codes and standards for Government approval prior to initiating preliminary design.

3.3.7 The contractor's design shall consider the phasing of construction to minimize impacts and/or outages to AEDC operations.

3.4 ENERGY AND ENVIRONMENTAL REQUIREMENTS

3.4.1 The contractor shall specify the use of the maximum practicable amount of recovered materials consistent with the performance requirements, availability, price reasonableness, and cost-effectiveness.

3.4.2 The contractor shall consider sustainability, renewability, energy conservation, pollution prevention, and waste reduction to the maximum extent practicable in developing construction design specifications.

3.4.3 Designs shall include all life cycle cost effective energy conservation systems/features IAW current Executive Orders, DoD Instruction 4170.11, Energy Independence and Security Act of 2007, and Guiding Principles for Federal Leadership in High Performance and Sustainable Buildings. Energy cost budgets shall be reduced by 30 percent for all new construction and 20 percent for all renovations as compared to American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE) 90.1 baseline building performance ratings. Submit alternate proposals, if appropriate, which would reduce energy consumption below the expected energy budget. Indicate the effect on facility construction and operation costs of any such proposals.

3.4.4 If a design contains energy-consuming products listed in the ENERGY STAR Program or Federal Energy Management Program (FEMP) the contractor shall specify products which meet ENERGY STAR and FEMP requirements and are in the upper 25 percent of energy efficiency in their class.

3.4.5 For products that consume power in a standby mode and are listed on FEMP's Low Standby Power Devices product listing, the contractor shall specify items which meet FEMP's standby power wattage recommendations. If FEMP has listed a product without a corresponding wattage recommendation, the contractor shall specify items which use no more than one watt in their standby power consuming mode. When it is impracticable to meet the one-watt requirement, the contractor shall specify items with the lowest standby wattage practicable.

3.4.6 Environmental Protection Agency (EPA) and US Department of Agriculture (USDA)-designated items: The contractor shall specify products containing recovered materials and bio-based products to the maximum extent practicable without jeopardizing the intended use of the product while maintaining a satisfactory level of competition at a reasonable price. The Product information is available at <https://www.usda.gov>, keyword: BioPreferred Program and <https://www.epa.gov>, keyword: Comprehensive Procurement Guideline Program.

3.4.7 Prohibited Products: The contractor shall not specify products or materials that contain lead, chromium, mercury, cadmium, silver, barium, selenium, beryllium, or arsenic except as expressly authorized by the Government. If no substitutes for products containing the listed materials are available, and the Government approves the use of products containing the listed materials, their exact location shall be highlighted and detailed on project drawings. The use of Class 1 ozone depleting substances (ODSs) are also prohibited.

3.4.8 Asbestos Products: The contractor shall not specify products or materials that contain asbestos except as expressly authorized by the Government. If no substitutes for asbestos products are available, and the Government approves the use of asbestos products, their exact location shall be highlighted and detailed on project drawings.

3.4.9 The contractor shall submit calculations for quantity (acres) of disturbed area resulting from anticipated construction activities.

3.5 DRAWINGS

3.5.1 General: The contractor shall provide design drawings which meet the requirements of the TO. Unless otherwise specified, drawings shall be submitted in electronic format in Size American National Standards Institute (ANSI) D (22 inches H x 34 inches W). In order to increase the value of pictorial presentation, proposed construction should be heavy lined, while existing items shall be light lined or ghosted. Fabrication drawings and manufacturer drawings are prohibited for use as contractor design drawings. Standard details not applicable to the project shall not be provided on design drawings. Details of standard products or items that are adequately covered by specifications shall not be included on design drawings. The contractor shall provide an index of TO drawing sheets in each submittal. Drawings presented “for information only” do not need to be included. Additional drawings shall be included as they are developed by the contractor and the index shall reflect the additional drawings. Mixed media formats will not be accepted. Files must be fully editable and match the drawings submitted. Referenced files shall be permanently bound to the drawing files. Partial files, files with missing attachments or layers or files formatted as read-only or in other ways protected will not be accepted.

3.5.2 Design drawings: Design drawings shall contain all necessary views, details, dimensions, notes, bill of materials, etc., which accurately and clearly convey all aspects of the contractor’s design. Design drawings shall contain sufficient detail for the Government to assess validity of the design and conformance with the TO and for the contractor’s fabricator(s) to generate fabrication drawings without need for further explanation from the designer. Design drawings and specifications shall be prepared in design submittals such that AEDC could construct the project without any additional assistance from the contractor. The contractor shall deliver drawing files that are fully compatible and editable by AutoCAD 2018 or later, as well as PDF versions generated from the AutoCAD drawings (photocopying is prohibited). The contractor shall use standard AEDC title blocks and borders on all design drawings, supplied by the Government. The contractor’s design drawings shall use US Customary units. Dimensions shall be in terms of inches or feet or a combination thereof as applicable. Elevations shall be drawn to a 1/8” = 1'-0” scale and other visual information as required. The site and exterior utility drawings shall use a minimum scale of 1" = 30' unless otherwise indicated. Additionally, the overall site plan for this project shall be on one drawing sheet.

3.5.3 Fabrication Drawings: The contractor shall electronically submit fully legible PDF format file(s). Fabrication drawings shall be created using a Computer-Aided Design and Drafting (CADD) program and the PDF file(s) electronically generated from the CADD file to maximize readability (photocopying is prohibited). Fabrication drawings shall be completely detailed following accepted drafting practice standards and contain all information necessary to fabricate the part(s). These drawings shall be organized and presented such that conformance with the contractor’s design drawings and the TO can be easily verified by the Government. The fabricator’s standard drawing title block and borders are acceptable. Fabrication drawings shall use US Customary units. Dimensions shall be in terms of inches or feet or a combination thereof as applicable.

3.5.4 Manufacturer drawings: The contractor shall electronically submit fully legible PDF format file(s). Even though CADD drawings are not required to be submitted, all drawings created by manufacturers for this project shall be created using a CADD program and the PDF file(s) electronically generated from the CADD file to maximize readability (photocopying is prohibited). Manufacturer drawings may use the International System of Units (SI) if the products were designed and fabricated in SI units, otherwise; manufacturer drawings shall be submitted in US Customary units.

3.6 SPECIFICATIONS

3.6.1 The specification format shall conform to the Construction Specifications Institute's (CSI's) Manual of Practice. Specifications shall consist of one editable electronic set of specifications in Microsoft Word (e.g., .doc or .docx file format) and one electronic set in PDF format.

3.6.2 Considerations for specifications include, but are not limited to:

- a) The use of small amounts of materials, which are not complicated in their application, does not require coverage in the written specifications provided descriptive information and applicable standards, if required, are noted on the drawings.
- b) References in specifications shall be to widely recognized standards or specifications promulgated by governments, industries, or technical societies. In all instances when there is an applicable specification or standard, a reference thereto, as well as the latest date of the standard, shall be used.
- c) Brand names may be used but shall be prefaced by the phrase "similar and equal to." When a "brand name or equal" description is necessary, specifications shall clearly identify and describe the particular physical, functional, or other salient characteristics of the brand-name items which are considered essential to satisfying the requirement.

3.7 SUBMITTAL REGISTER

The contractor shall develop submittal requirements required during construction as part of the design phase of the contract. This shall be accomplished by providing a contractor submittal register to the Government during design. A submittal register shall be attached to each section of the specifications for the submittal requirements of that section. The contractor shall be responsible for listing all required submittals necessary to ensure the project requirements are met. The register shall identify submittal items such as shop drawings, manufacturer's literature, certificates of compliance, material samples, guarantees, test results, etc., that the contractor shall submit for review and/or approval action during the life of the construction contract. The contractor shall place all the submittal register pages in an appendix of the final specifications.

3.8 COORDINATION AND EXECUTION OF WORK

3.8.1 Construction may not begin until the final design documents are approved by the Government. Once the design is approved, design changes are not allowed without Government approval. In some cases, fast-tracked design/build projects (projects for which construction begins at a mutually agreed point prior to completion of the design and construction drawings) may be required and will be clearly defined in the TO with various fast-tracked phases. In case

of a fast-tracked project, construction may start only upon completion and approval of the design phases as defined in the TO.

3.8.2 Record Drawings: Submit drawing requests to the Government Representative 14 days prior to need date. The Government Representative will then coordinate the request.

3.8.3 Site Investigations: The contractor is responsible for determining existing site conditions and coordinating new work with existing conditions, IAW Federal Acquisition Regulation (FAR) 52.236-3, *Site Investigation and Conditions Affecting the Work*. Site investigations shall be sufficiently thorough to ensure design details are compatible with the project site. Necessary arrangements for site investigation visits to AEDC shall be made through the Government Representative.

3.8.4 Government Representative: The Government will designate a representative to perform technical surveillance. This representative will recommend to the CO or COR acceptance or rejection of work under this contract. All references to the "Government Representative" within the specification or other project documents shall mean the AF Project Manager/Engineer or the AEDC Operating Support contractor personnel performing technical surveillance on behalf of the Government. Government technical surveillance does not relieve the contractor from conducting its own surveillance.

3.9 DESIGN SUBMITTAL REQUIREMENTS, REVIEWS, AND MEETINGS

Design submittal requirements for individual units of work are specified in the TO and should comply with the following requirements for each type of submittal. The contractor shall electronically submit each submittal in a file type and media format defined by the Government in the TO unless otherwise specified.

3.9.1 Within 21 days after NTP, the contractor shall submit a complete design schedule with all submittals and review times indicated by calendar dates for approval by the Government. If the contractor does not meet a scheduled design submittal date without notifying the Government in writing one week in advance, seven days will be added to the Government review period.

3.9.2 Design submittal requirements are described in Attachment 2. The TO will identify tailoring of design submittal content, if necessary. Design submittals shall be made at the 30 percent, 60 percent, 90 percent, and final design stages or as stated in each TO.

3.9.3 Required documents shall be submitted 21 calendar days prior to a scheduled review meeting or IAW the schedule given in the specific instructions for each TO. The contractor shall electronically submit documents in a file type and media format defined by the Government in the TO. The time required by the Government to review submittals made by the contractor under this contract will be negotiated on each TO.

3.9.4 The contractor shall present design packages as part of design reviews. The reviews may be tailored, as noted in each TO. Unless otherwise noted, design reviews shall be conducted at the AEDC location of performance. It is the contractor's responsibility to schedule and coordinate attendance of all personnel that developed the design submittal at each review. The designers for each discipline shall attend all design reviews.

3.9.5 After each review of the design package submittals, the contractor shall make changes and corrections as necessary. The contractor is cautioned that if it believes the action required by any comment exceeds the requirements of the TO, it should take no action and notify the CO in writing immediately. All review comments for each submittal shall otherwise be incorporated into the following design submittal. The contractor shall supply the CO written disposition of the review comments within 14 calendar days of receipt of the comments.

3.9.6 The contractor shall prepare and submit minutes for each review or meeting. Minutes shall be submitted to the CO or their representative within five calendar days after the meeting.

3.9.7 Government personnel may visit the contractor's office to review progress of projects.

3.9.8 During each project design, the Government retains the right to continue to provide input and finalize the concept up through the final design submittal.

3.9.9 Government approval will not relieve the contractor of the responsibility for any error, which may exist, as the contractor is responsible for the design and construction of all work.

3.10 COST ESTIMATING GUIDANCE

This section applies only to TOs that are capability design only or the first phase of a two-phase design/build effort.

3.10.1 The Designer/Cost Estimator shall follow the specified guidance below if either of the following criteria are met:

- a) The project scope is considered to be REAL PROPERTY as defined by the Real Property Officer and/or Programing Chief and the construction cost estimate, to include only funded cost, is greater than \$1,000,000.
- b) If the Government designates that all or a portion of the cost estimate must be delivered utilizing the below guidance.

3.10.2 The Designer/Cost Estimator shall develop a project construction cost estimate following the below guidance when either of the above criteria is met.

3.10.3 The estimate shall utilize the most current DoD cost data (Cost Book) by preparing the estimate in current versions of either Parametric Cost Engineering System Software (PACES) or Micro-Computer Aided Cost Estimating System 2nd Generation (MII) software IAW Unified Facilities Criteria (UFC)-3-740-05.

3.10.4 The estimate shall be developed by experienced personnel. Specifically, the lead estimator shall have a minimum of five years experience estimating Government construction.

3.10.5 Consistent with AF policy and business rules, the estimate shall be developed with a methodology (minimum parametric) and estimate class that is suitable for the respective design maturity as determined from Association for the Advancement of Cost Engineering International (AACEI) Recommended Practice 56R-08. Per AF policy, a design must reach 35 percent maturity before considered an AACEI Class 3 estimate.

3.10.6 The estimate shall be well-documented. All estimator assumptions, notes, and quantity calculations shall be documented within the software note fields, a Basis of Estimate (BOE) document, backup sheets, or a combination of methods.

3.10.7 IAW UFC-3-740-05, the estimate shall be formatted with the Uniformat II Work Breakdown Structure (WBS) for military projects as provided in the PACES and MII software.

3.10.8 The estimate shall accurately capture all project features associated with the respective design. The Designer/Cost Estimator shall demonstrate quantity validity and estimate traceability by documenting or recording quantity calculations (or quantity assumptions) in quantity backup sheets, separate estimator notes, within the estimating software notes fields or a combination of methods.

3.10.9 Per UFC 3-740-05, Designer/Cost Estimator shall utilize allowances and/or design contingencies to cover construction cost increases due to design incompleteness, detail changes, alternative design changes, and associated costing inaccuracy. Design contingencies will normally decrease, as design information becomes known. Usage of all assumptions, allowances and/or design contingencies shall be fully documented and justified within the BOE or estimate software note fields.

3.10.10 The Designer/Cost Estimator shall tailor the estimate to the specific geographic location. If utilizing PACES, the Designer/Cost Estimator shall apply the appropriate Area Cost Factor, as required by UFC-3-701-01. If utilizing MII, the DESIGNER/COST ESTIMATOR shall utilize project area specific labor and equipment libraries.

3.10.11 The Designer/Cost Estimator shall inflate DOD cost data (labor, equipment, and material) to current year value through defensible means. Additionally, the Designer/Cost Estimator shall escalate project pricing to mid-point of construction by applying the Naval Facilities Engineering Systems Command (NAVFAC) Building Cost Index (BCI). Usage and calculations of inflationary/escalatory factors shall be documented in the BOE and/or software notes.

3.10.12 The Designer/Cost Estimator shall adjust or account for DOD data productivity rate changes, if appropriate, to job site specific requirements. Requirements such as, safeguards, security, safety, confined space, double handling of materials, restricted laydown space, unusually large or small construction work limits and other constructability issues shall be considered in determining proper productivity. The derivation of productivity factors shall be well documented and justified within the BOE and/or software notes.

3.10.13 The Designer/Cost Estimator shall fully document within the BOE or software notes all user defined cost elements that differ from the current DOD Cost Book.

3.10.14 Labor availability issues shall be considered and an assumption shall be documented. If applicable at project location, the estimate shall include an assumed amount of labor premiums, subsistence, or both.

3.10.15 The estimate shall be prepared utilizing current acquisition strategy or assumed strategy with the estimate and associated documentation clearly identifying prime and subcontract tiers to facilitate compounding calculations. Additionally the Designer/Cost Estimator shall apply indirect costs (markups) as appropriate and fully document the basis of each in the BOE or software notes fields. Examples of markups to consider are, but not limited to:

- a) Direct Cost Markups – Productivity, Overtime, Labor Premiums, Sales Tax, Misc, Payroll Tax, Workman’s Comp, Inflation (to current year)

- b) Subcontractor Markups – Job Office Overhead (JOOH), Home Office Overhead (HOOH), Profit, Bond, Excise, Miscellaneous
- c) Prime Contractor Markups – JOOH, HOOH, Profit, Bond, Excise Tax, Miscellaneous
- d) Project Markups – Escalation (project mid-point), Design Contingency, Construction Contingency (Program Specific), Supervision, Inspection and Overhead (SIOH), Miscellaneous, Special (such as Gross Receipts Tax)

3.10.16 IAW AF policy and business rules, the Designer/Cost Estimator shall develop a BOE narrative report. The BOE is a document, developed by the estimating team, which describes how an estimate is prepared and defines information used in support of development. The BOE shall provide traceability of underlying estimate characteristics and provide enough detail to enable an independent, proficient cost engineer to recreate the estimate using only the details provided.

3.10.17 The Designer/Cost Estimator may utilize the BOE template developed and maintained by the Air Force Civil Engineering Center. The most current version will be provided by the Government and can be found at the below link.
<https://usaf.dps.mil/teams/10758/CEIPRegions/Shared%20Documents/Forms/AllItems.aspx?InitialTabId=Ribbon%2ERead&VisibilityContext=WSSTabPersistence>

3.10.18 The Designer/Cost Estimator may utilize the estimating software note fields to document many of the BOE requirements. If PACES or MII notes fields are utilized, then the Designer/Cost Estimator may simply reference them in the BOE document. Designer/Cost Estimator shall provide a detailed report “with notes” extracted from the software (PDF format) to facilitate BOE informational review to individuals without software access.

3.10.19 This estimate is subject to review, validation, and certification by an appointed independent AF Cost Engineering Reviewer/Approver (R/A) prior to leadership approval of the baseline Programmed Amount (PA). The Designer/Cost Estimator shall respond and reconcile all R/A comments in writing. If the Designer/Cost Estimator concurs with the comments, the R/A recommendations shall be incorporated into the estimate by the Designer/Cost Estimator, including supporting documentation, then re-submitted for back-check. If the Designer/Cost Estimator non-concurs with R/A comments, then the Designer/Cost Estimator shall provide written response detailing the reason for non-concurrence. If requested by the Designer/Cost Estimator, the installation will facilitate a meeting with R/A to discuss estimate comments to aid in timely reconciliation.

3.10.20 Deliverables:

3.10.20.1 The Designer/Cost Estimator shall submit native estimate file for PACES (.paces) or MII (.mlp)

3.10.20.2 The Designer/Cost Estimator shall submit BOE narrative as described in 1.2 above

3.10.20.3 The Designer/Cost Estimator shall submit summary level estimate report from software to assist installation with populating the DD Form 1391 (PDF format).

3.10.20.4 The Designer/Cost Estimator shall submit detailed estimate report with notes from software (PDF format).

3.10.20.5 The Designer/Cost Estimator shall submit all estimator backup to include takeoff sheets, backup calculations, quotations and any other information that demonstrates traceability, defensibility, and reasonableness of the estimate. The documents may be attached to the BOE or bundled separately.

3.10.21 Applicable References

- a) UFC 3-701-01 – *DoD Facilities Pricing Guide*
- b) UFC 3-730-01 – *Programming Cost Estimates for Military Construction*
- c) UFC 3-740-05 – *Construction Cost Estimating*
- d) AACEI 34R-05 – *Basis of Estimate*
- e) AACEI 56R-08 – *Cost Estimate Classification System – As Applied in Engineering, Procurement, and Construction for the Building and General Construction Industries*

SECTION 4

4 CONSTRUCTION

4.1 SCOPE

This section outlines the required services to be provided by the contractor for the construction phase of a TO under this IDIQ contract. The contractor shall provide sufficient technical, supervisory, and administrative personnel to ensure accomplishment of the work specified as required for the duration of the contract. PE services may also be required during construction for consultation. PE services, if required, will be specified in the TO requirements document.

4.2 DEFINITIONS

Certain terms used in the contract documents are defined below. Definitions and explanations contained in this section are not complete, but are general for the work to the extent that they are not stated more explicitly in another element of the contract documents.

4.2.1 Furnish. The term "furnish" means to supply and deliver to the project site, ready for unloading, unpacking, assembly, installation, and similar operations.

4.2.2 Install. The term "install" describes operations at the project site, including the actual unloading, unpacking, assembly, erection, placing, anchoring, applying, working to dimensions, finishing, curing, protecting, cleaning, and similar operations.

4.2.3 Provide. The term "provide" means "provide complete in place"; that is, "furnish and install."

4.2.4 Postconsumer material (from FAR clause 52.223-9): "[A] material or finished product that has served its intended use and has been discarded for disposal or recovery, having completed its life as a consumer item. Postconsumer material is a part of the broader category of 'recovered material'."

4.2.5 Recovered material (from FAR clause 52.223-9): "[W]aste materials and by-products recovered or diverted from solid waste, but the term does not include those materials and by-products generated from, and commonly reused within, an original manufacturing process."

4.3 JOB CONDITIONS

4.3.1 The contractor shall furnish new components and devices, complete, with all necessary materials for installation to meet this objective.

4.3.2 The contractor shall keep the work area clean and free from waste and debris. Dispose of waste as directed by the Government Representative. An approved construction/demolition waste disposal landfill is located about two miles from the main industrial area. This landfill is approved for surplus earth, stone, broken concrete and building materials waste. Building materials waste also includes containers, dunnage, and other related items. A separate asbestos waste disposal landfill is also located about two miles from the main industrial area. The contractor performing asbestos abatement/removal and disposal may utilize this landfill, but it must be in strict compliance with all rules and regulations.

4.3.3 The contractor shall be responsible for protection of materials and equipment stored at the job site.

4.3.4 The contractor shall not close or obstruct egress width to any building or site exit without written permission as coordinated through the Government Representative.

4.3.5 The contractor shall not disable or disrupt building fire or life safety systems without written permission as coordinated through the Government Representative.

4.3.6 The contractor shall conform to applicable procedures when any hazardous or contaminated materials are discovered.

4.3.7 In the case of any unsafe conditions, the contractor shall cease operations and immediately notify the Government Representative. Do not resume operations until notified by the Government Representative.

4.3.8 The contractor shall provide temporary barricades and other forms of protection as required to protect personnel from injury due to the work.

4.3.9 The contractor shall provide protective measures as required to provide free and safe passage of personnel to and from occupied portions of the building. Provide temporary signage as needed or directed.

4.3.10 The contractor shall not close, block, or otherwise obstruct streets, walks, or other occupied or used facilities without coordination through the Government Representative.

4.3.11 All scaffolding shall be IAW AEDC Safety Standard D10.

4.3.12 Available utilities

4.3.12.1 Water: All reasonable required amounts of potable and raw water will be made available without cost to the contractor from existing systems, outlets, and supplies near the vicinity of the work. All water shall be carefully conserved. All temporary connections and piping installed by the contractor shall be removed in a manner satisfactory to the CO prior to final acceptance of the work.

4.3.12.1 Electricity: All reasonably required amounts of 60 HZ electric power at 110V - 480V will be made available without cost to the contractor from existing electrical systems near the vicinity of the work. At no cost to the Government, the contractor shall install and maintain any necessary supply connections and facilities in a manner satisfactory to the CO. All electrical service shall be removed in a manner satisfactory to the CO.

4.3.13 Staging/storage sites/on-site offices

4.3.13.1 Staging/storage site locations shall be approved by the CO prior to use. The contractor shall submit written request(s) for approval of the location(s) being requested a minimum of 14 calendar days prior to use. The contractor shall store all supplies and equipment on staging/storage and construction sites so as to preclude theft or damage. In addition, the contractor shall maintain sites in a neat and orderly manner. Protection and security for materials and equipment on sites is the sole responsibility of the contractor.

4.3.13.2 The contractor shall request the CO's approval of temporary storage units/boxes, office/storage trailers, and dumpsters. The contractor shall not bring any such items to the job site

without the CO's approval. A nameplate shall be visually displayed on the exterior of all storage units with the following information: Contractor's company, name of contractor's POC(s) and telephone number(s) where he/she can be reached, (during and after normal working hours) and contract number.

4.3.14 The contractor shall provide toilet facilities in a ratio of not less than one for each 30 persons or fraction thereof for each worksite. Connections to existing sanitary sewer manholes may be made. The contractor shall supply drinking water from connections to the existing potable water system. The contractor shall take proper precautions to protect the water system from damage and contamination. Any temporary connections made for drinking water or toilet facilities shall be removed by the contractor in a manner satisfactory to the CO/COR before final acceptance of the work.

4.4 PROTECTION OF EXISTING IMPROVEMENTS

4.4.1 Protection of the grounds, walkways and structures from deterioration due to excessive vehicular traffic, equipment disturbance and material staging is critical to maintaining the quality appearance of AEDC. The contractor shall protect against damage incurred by service vehicles, construction activities, maintenance and repair, and all other work that detracts from the overall appearance of the grounds and increases the costs of grounds maintenance.

4.4.2 Prior to commencement of work, the contractor shall inspect areas in which work will be performed. File a report with the CO listing existing conditions which could be construed as damage resulting from the work. AEDC representatives may also photograph or video tape areas or improvements susceptible to damage to document conditions prior to the start of work.

4.4.3 The activity of vehicles and equipment on the grounds shall be limited to only that which is essential and required for completing the work. The Government Representative will approve all access and haul routes required for the work prior to any on-site operations to preclude damage to sensitive utilities or improvements which may not be readily apparent.

4.4.4 The contractor shall minimize travel paths and staging areas to limit the impact and damage to work areas. Lay out the work site, material and equipment staging, and vehicle and equipment activity to minimize visibility and aesthetic impact of activities to surrounding areas.

4.4.5 The contractor shall protect above and below grade utilities, plant life, lawns and other features or improvements to remain.

4.4.6 The contractor shall protect benchmarks, existing structures, sidewalks, curbs and pavements from excavation equipment and vehicular traffic.

4.4.7 The contractor shall use protective materials, such as sheeting, steel plating, rigger pads, to protect grounds and pavement from heavy equipment activity and damage.

4.4.8 The contractor shall implement a daily procedure for area clean up and organized parking of vehicles and equipment.

4.4.9 The contractor shall provide temporary barricades, shoring, bracing, supports, or other forms of protection as required to protect all existing improvements not scheduled to be demolished for both above and below ground improvements and to prevent movement, settlement, collapse, or damage of any kind.

4.4.10 The contractor shall maintain personnel and vehicular access and existing utilities to all buildings and other structures during the course of the work. Any changes or interruptions shall be coordinated and approved by the Government Representative well in advance. Temporary protections or services may be required as a condition of approval of any such change or interruption.

4.4.11 The contractor shall be responsible for damage caused by the work and shall repair or replace same to the satisfaction of the Government. Obtain the Government Representative's approval of methods, materials, equipment or products for repair or replacement prior to use. All such repair or replacement shall be performed promptly at no additional cost to the Government.

4.4.12 The contractor shall remove all temporary barricades, shoring, bracing, supports, or other forms of protection at completion of work. Restore all adjacent structures and surfaces to conditions existing prior to construction. Clean up all stockpiles and spoil areas and restore all surrounding areas to conditions existing prior to construction.

4.5 TOOL CONTROL

The contractor shall ensure tool control is followed IAW AEDC Instruction (AEDCI) 21-3, Tool Control. Applicability of this instruction will be identified, as necessary, in each TO.

4.6 EPA-DESIGNATED ITEMS

The contractor, upon completion of TO IAW FAR 52.223-9, shall:

4.6.1 Estimate the percentage of the total recovered material content for EPA-designated item(s) delivered and/or used in contract performance, including, if applicable, the percentage of post-consumer material content; and

4.6.2 Submit this estimate to the CO.

4.7 ELIMINATION OF ODS CHEMICALS

The use of Class I ODSs is prohibited. Reference text of Department of the Air Force Federal Acquisition Regulation Supplement (DAFFARS) clause 5352.223-9000 (October 2024).

4.8 DISALLOWED PRODUCTS

Do not use products or materials listed on the banned chemicals/ingredients within AEDC E06 Safety, Health, and Environmental (SHE) Standard for Hazardous Materials on this project except as expressly authorized by the CO. If no substitutes for products containing the listed materials are available, and the CO approves the use of products containing the listed materials, highlight and detail their exact location on the drawings.

4.9 LEAD BASED PAINT ABATEMENT

When required by the TO, check for lead-based paint and develop project documents for abatement and/or encapsulation. Take adequate number of paint samples and test for the presence of lead-based paint. The contractor shall submit results of lead-based paint sampling to the Government for verification and record keeping purposes. Where presence of existing lead-based paint will have an impact on project accomplishment, identify location and quantity of such paint on drawings. Develop specifications and drawings that indicate method of either

abatement or encapsulation. Ensure specifications incorporate the most recent requirements and changes of subpart D of 29 Code of Federal Regulation (CFR) part 1926.

4.10 ASBESTOS

4.10.1 Products: Do not use products or materials that contain asbestos on this project except as expressly authorized by the CO. If no substitutes for asbestos products are available, and the CO approves the use of asbestos products, highlight and detail their exact location on the drawings and identify their location in the field following 29 CFR 1926.1101 guidelines.

4.10.2 Abatement: Due to the age of facilities, asbestos may be present in many projects executed under this contract. If specific asbestos requirements are known, they will be identified in the TO. Site investigations should be conducted to determine extent of asbestos abatement required IAW SOW 3.8.3. The Occupational Safety and Health Administration (OSHA) has set standards at 29 CFR 1926.1101 for exposure to airborne concentrations of asbestos fibers, methods of compliance, medical surveillance, housekeeping procedures, and measures that must be taken when working with or around asbestos-containing other materials. 29 CFR 1926.1101 has been identified as applicable to construction (29 CFR 1926.55 gases, vapors, fumes, dusts, and mists). The EPA has established standards at 40 CFR 61.140-157 for the control of asbestos emissions to the environment and the handling and disposal of asbestos wastes.

4.11 HAZARDOUS WASTE

Where hazardous waste (as identified in 40 CFR 261) is generated, follow the procedures in AEDC Safety Standard E18, Managing Wastes Containing Chemical or Petroleum Products, for storing and turning in hazardous waste. These procedures include the requirement to complete the Waste Identification Form and the Chemical Waste Data Sheet, or equivalent, which will be provided by the Government Representative.

4.12 HAZARDOUS MATERIAL IDENTIFICATION AND MATERIAL SAFETY DATA

The following procedures shall be followed to meet 40 CFR 370, EPA Hazardous Chemical Reporting and Community Right to Know Requirement.

4.12.1 Procedures:

4.12.1.1 The contractor shall supply information on the hazardous materials it brings on AEDC property prior to beginning on-site work. Hazardous materials may include, but are not limited to: solvents, paints, adhesives, and acids.

4.12.1.2 The information required is:

- a) Company name, POC, and phone number
- b) Brief statement indicating how the hazardous material will be used within the scope of the contract
- c) List of all hazardous materials to be used (product name, manufacturer's name, and address)
- d) Amount of each material to be stored on-site and where it will be stored
- e) Where the product Safety Data Sheets (SDS) will be maintained

4.12.1.3 All unused product is the responsibility of the contractor and shall be removed from AEDC property at the completion of the project.

4.12.1.4 The contractor shall coordinate with the Government Representative and the Hazardous Waste Operations Group (AEDC.WasteTurnIn@us.af.mil) if any hazardous waste is to be generated.

4.12.2 Definitions:

4.12.2.1 Hazard communication standard. A chemical right-to-know law under OSHA that requires chemical manufacturers and importers to assess the hazards of chemicals they make or import and to distribute this information to inform workers of the hazards associated with these chemicals. This written information is a SDS.

4.12.2.2 Hazard classes. Hazardous materials that have been grouped into classes by the Department of Transportation. These classes include explosives, flammables, oxidizers and organic peroxides, compressed gases, corrosives, and poisons.

4.12.2.3 Hazardous material. Any substance that may be harmful when used. Specific substances have been designated as hazardous under the Clean Water Act, the Resource and Conservation and Recovery Act, and the hazardous air pollutants under the Clean Air Act.

4.12.2.4 Hazardous waste. Any waste that may cause or significantly contribute to serious illness or death or that may pose a substantial threat to human health or the environment, if not properly managed. Hazardous wastes may be solids, liquids, semi-solids, or compressed gases.

4.13 WELDING/BRAZING REQUIREMENTS

4.13.1 Welding/Brazing Requirement shall meet American Society of Mechanical Engineers (ASME) Section IX & ASME Section VII, Div. 1.

4.13.2 Submittals. Prior to beginning fabrication, the contractor shall submit the following IAW ASME Section IX & ASME Section VIII, Div. 1 when applicable:

4.13.2.1 Welding Procedure Specifications(s) (WPS) & Procedure Qualification Record(s)

4.13.2.2 Welder and/or Welding Operator Performance Qualification(s) (WPQ) that demonstrate the welders are qualified to perform the provided WPS(s).

4.13.2.3 Provide documentation supporting welder continuity in maintaining their qualifications.

4.13.2.4 Brazing Procedure Specification(s) & Procedure Qualification Record

4.13.2.5 Any drawings detailing the work that is to be performed

4.13.2.6 Drawings must state to which fabrication/design code to which the construction is to be fabricated and/or inspected

4.13.2.7 Inspection plans

4.13.2.8 Heat treatment procedures

4.13.2.9 Provide documentation for U or R stamps when applicable

4.13.3 Post fabrication inspection reports shall include, as applicable: Visual Inspections, Radiography, Magnetic Particle, Vacuum Leak Test, Pressure Tests, Heat Treat Test Report, Liquid Penetrant, Initial Service Leak Test

4.13.4 In the event that welding/brazing is required to close a confined space, prior to closing the space, the contractor shall provide 24 hour notice of closing to allow for the Government representative to inspect the interior of the space.

4.14 EXCAVATION REQUIREMENTS

The location of existing underground utilities as shown on the contract drawings is specified in as-built drawings. There may be underground utilities which are not shown on the contract drawings and these utilities may be dangerous if ruptured. If the contractor unearths or strikes any underground utility not shown, the excavation work shall cease and the CO and Government PM shall be notified immediately. With caution, the contractor shall determine the exact location of such utilities and will not proceed with further excavation unless otherwise advised by the CO.

4.14.1 All excavation will require a dig permit.

4.14.2 The contractor shall be responsible for keeping all subsurface utility markings and flagging up once AEDC has marked the worksite.

4.14.3 The contractor shall utilize Hydrovac to uncover utilities in congested industrial areas.

4.14.4 The contractor shall hand dig within four feet of all marked utilities.

4.14.5 Wall penetrations require a dig permit prior to penetration.

4.14.6 Excavators shall not have teeth on the bucket to prevent damage to utilities.

4.14.7 Any damage to utilities shall be reported to the CO and Government PM immediately.

4.15 PAINTING REQUIREMENTS

4.15.1 The contractor shall accomplish all painting IAW Society for Protective Coatings (SSPC) Guide 6, SSPC PA 2, SSPC SP1, SSPC SP2 and SSPC SP3.

4.15.2 The contractor shall coordinate all painting activities with the Government representative prior to work being accomplished.

4.15.3 The contractor shall take precaution to prevent overspray including but not limited to physical barriers, working during unfavorable weather conditions, and installing signage as defined in SOW 4.15.4.

4.15.4 Signage

4.15.4.1 The contractor shall provide all sign stands and barricades necessary to post warnings along roads adjacent to the work area and at the entrances to all parking lots, through streets, and roads within a 500-foot perimeter around the work area. Signs shall be at least two feet by two feet and weighted for stability.

4.15.4.2 Signs shall provide the following information: NOTICE: SPRAY PAINTING IN PROGRESS, PASS AT OWN RISK". The signs shall be maintained by the contractor.

4.15.4.3 Signs shall have a white background. The word “NOTICE” shall be 4-inch letters; white on blue, balance of words shall be 3-inch letters, black on white.

4.15.4.4 After all signage is in place, the contractor shall notify the Government representative who will verify the signage has been installed per specifications and provide approval to commencing painting.

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SECTION 5

5 TO ISSUANCE PROCEDURES

As requirements are identified, the Government will issue a FOPR that will describe the services required along with the response time for the TO proposal. A site visit may be scheduled with the contractor(s) to verify measurements and elements of work. Procedures for issuing TOs are stated as follows:

5.1 GENERAL

5.1.1 Each basic contract award made as a result of this solicitation will cite a unique contract number. Individual TOs will be placed under each basic contract.

5.1.2 Orders will be placed on a DD Form 1155, Order for Supplies or Services.

5.1.3 The Government will not be obligated to reimburse the contractor for work performed or any costs incurred, nor shall the contractor be obligated to perform or otherwise incur costs except as authorized by executed TOs.

5.1.4 TOs will be FFP and clearly define the scope of work to be performed or the performance desired. Each TO will contain applicable provisions and the specification package required for that project.

5.1.5 TO evaluation methods may include LPTA or tradeoff. Each TO will state the method of evaluation.

5.1.6 TO projects may be non-complex, performance-oriented tasks requiring no design, minimal design, or may be complex construction requiring design development for design/build construction, or a combination thereof.

5.1.7 Contract holders are expected, in good faith, to participate in the TO competitions. In the event a FARM awardee is unable to submit a proposal on a particular FOPR, the contractor is required to notify the CO in writing within five business days from receipt of the FOPR of the reason(s) for non-participation.

5.1.8 All terms and conditions identified in the basic contract are in full force by reference for each TO.

5.2 COMPETITION

5.2.1 Competition for TOs is limited to those awardees under this contract. All awardees will be given a fair opportunity to be considered for each TO. Upon determining the need to issue a TO, all awardees will be considered equally against the stated criteria.

5.2.2 Unless the CO applies the exceptions noted below, each TO will be awarded as a result of competition. The CO's decision as to the selection for award of a TO cannot be protested unless the protest is on the grounds that the order increases the scope, period, or maximum value of the contract or the value of the order is in excess of \$25 million.

5.2.3 Each contractor shall be given a fair opportunity to be considered for award of a TO. The CO reserves the right to make award of a TO without competition based upon one of the circumstances described below:

5.2.3.1 The agency need for the supplies or services are so urgent that providing a fair opportunity would result in unacceptable delays;

5.2.3.2 Only one awardee is capable of providing the supplies or services required at the level of quality required because the supplies or services ordered are unique or highly specialized;

5.2.3.3 The order must be issued on a sole-source basis in the interest of economy and efficiency as a logical follow-on to an order already issued under the contract, provided that all awardees were given a fair opportunity to be considered for the original order; or

5.2.3.4 It is necessary to place an order to satisfy a minimum guarantee; or

5.2.3.5 A statute expressly authorizes or requires that the purchase be made from a specific source.

5.3 PREPARATION OF OFFERS

5.3.1 Offerors are expected to participate in site visits as well as read and understand the requirements document, drawings, specifications, schedule, and all instructions. Failure to do so will be at the offeror's risk.

5.3.2 Each offeror shall furnish the information required by the TO FOPR.

SECTION 6

6 ON/OFF RAMP PROCEDURES

6.1 The Government intends that each contractor is available at all times to achieve effective competition on all TOs.

6.2 Contract holders are expected, in good faith, to participate in the TO competitions, and if successful, to deliver quality projects on time and on cost, to include management and supervision of subcontractors. The contractor is required to notify the CO in writing within five business days from receipt of the FOPR of the reason(s) for non-participation. The Government will evaluate all contractors' participation rates, reason(s) for non-participation, and performance evaluations for TOs completed or in effect. Failure to participate in good faith and/or an overall annual performance assessment rating of Marginal or Unsatisfactory may result in the contractor being off-ramped from the multiple award program. An off-ramp decision results in the contract being terminated.

6.3 In the event contractor(s) are off-ramped within 36 months of initial contract award, the Government reserves the right to on-ramp contractor(s) to maintain a pool of up to eight contractors. On-ramp contractors will be selected based on initial evaluation results from the basic multiple award solicitation in order of ranking. Contractor(s) must be active in the System for Award Management (SAM) and otherwise in compliance with the original multiple award solicitation. If found in compliance, on-ramp contractors may be offered a bilateral IDIQ contract with original guaranteed minimum and allowed to compete on future TO projects.

6.4 The Government will not use on-ramp procedures beyond the 36-month mark from award date of the original contracts.

SECTION 7

7 MANDATORY POST AWARD CONFERENCE

7.1 The contractor shall furnish to the Government one post award conference. The post award conference shall serve to meet contract minimums and serve as the first TO award to establish a contractual relationship with the Government.

7.2 The post award conference shall consist of a site visit one day in duration. The agenda may include contractor orientation, kick-off meeting, and base tour.

7.3 The post award conference shall include travel to and from Arnold AFB TN to include all per diem for contractor personnel in attendance.

7.4 The post award conference shall include fully burdened labor costs of contractor personnel in attendance.

7.5 Post-Award Conference Travel: For travel to Arnold AFB TN cost shall not exceed per diem rates as identified in <https://www.defensetravel.dod.mil/site/perdiemCalc.cfm> for Coffee County, TN. (See FAR 31.205-46, *Travel Costs*)

7.6 Transportation: If driving by company vehicle, driving rate is based on Internal Revenue Service (IRS) established mileage rate at <https://www.defensetravel.dod.mil/site/faqmileage.cfm>. Per FAR 31.205-46, if flying on a common US commercial carrier, "airfare costs in excess of the lowest priced airfare available to the contractor during normal business hours are unallowable except when such accommodations require circuitous routing, require travel during unreasonable hours, excessively prolong travel, result in increased cost that would offset transportation savings, are not reasonably adequate for the physical or medical needs of the traveler, or are not reasonably available to meet mission requirements. However, in order for airfare costs in excess of the above standard airfare to be allowable, the applicable condition(s) set forth above must be documented and justified." Rental car or company vehicle must be justified as more advantageous to the Government.

SECTION 8

8 WORK SECURITY

The contractor shall be required to comply with all security directives and other security requirements identified in the respective contracts. These requirements apply to all contractors on Arnold AFB in the performance of their contracts. Changes to these requirements must be coordinated with AEDC/TSD-IP, AEDC/TSD-SF, and/or AEDC/XPT, as appropriate.

8.1 DoD Credentials: The contractor shall ensure base access and circulation control credentials required for contract performance are obtained for all employees. The contractor shall retrieve all identification media upon completion or termination of the contract and from employees who depart for any reason before the contract expires (e.g., the employee quits, is terminated, retires, etc.).

8.1.1 The contractor shall immediately notify the issuing office for an appropriate update in the information system of record when a contractor employee refuses to return issued access media. If a local access credential is not returned, the Installation Commander may consider debarring the individual. All contractor personnel must immediately report a lost or stolen access credential to the local issuing office so the card can be updated as lost/stolen in the system of record.

8.2 The contractor shall comply with all installation access requirements as directed in DoD Manual (DoDM)-5200.08_Volume 3_AF Manual (AFMAN) 31-101_Volume 3, *Installation Perimeter Access Control*, as supplemented and Arnold AFB (AAFB) 31-101, *Integrated Defense Plan*. The AEDC Commander reserves the right to refuse entry and/or re-entry of any person to Arnold AFB for just cause to protect personnel, facilities, and property under their control when such actions are deemed necessary.

8.2.1 All persons performing on the contract will be required to provide documents to positively prove their identity prior to being vetted for base entry. Identification must be compliant with the REAL ID Act of 2005. Individuals who do not possess a driver's license or identification card compliant with the REAL ID Act and do not possess an acceptable alternate identification credential, must be escorted or denied access.

8.3 Freedom of Information Act (FOIA): The contractor shall comply with FOIA requirements IAW DoDM 5400.07, *DoD Freedom of Information Act (FOIA) Program*. Requests for information received by the contractor and covered by FOIA shall be forwarded to the AEDC FOIA office for processing.

8.4 Controlled Unclassified Information (CUI) shall be safeguarded as specified in DoDI 5200.48, *Controlled Unclassified Information (CUI)*. This may include information with legacy For Official Use Only (FOUO) markings, previously protected under DoDM 5200.01- V4, *DoD Information Security Program: Controlled Unclassified Information*. Sensitive information is information that if lost, misused, compromised or accessed by unauthorized personnel could adversely affect US national interest, the conduct of DoD programs, or the privacy of DoD personnel. It comes in many forms; personal data under the Privacy Act, personnel records, privileged data (chaplain and judge advocate records), investigative data, scientific and technical

information (STINFO), export-controlled data (critical technologies), proprietary data, or other CUI data. For general CUI guidance, contact AEDC/ IP at (931) 454-3290.

8.4.1 If access to CUI/Export Control/STINFO information is required, the contractor shall be required to be registered as a certified Government contractor through the Joint Certification Office, Defense Logistics Information Service (DLIS). To request DLIS Certification, submit a DD Form 2345, *Militarily Critical Technical Data Agreement*, to the DLIS, Battle Creek, MI address indicated at the top of the DD Form 2345 along with a copy of the company's state business license, incorporation certificate, sales tax identification form, or other documentation, which verifies the legitimacy of the company. A copy of their DD2345, *Militarily Critical Technical Data Agreement*, must be provided when issued. Except as allowed by Defense Federal Acquisition Supplement (DFARS) 252.225-7048, *Export-Controlled Items*, foreign nationals shall not have access to export controlled information without an export license. All export control laws as stipulated in both Export Administration Regulation and International Traffic in Arms Regulation must be followed without exception. For guidance in applying appropriate export control markings, contact AEDC/ XPT at (931) 454-5681.

8.4.2 The contractor must obtain Foreign Disclosure Office (FDO) approval through the CO prior to allowing any non-US or un-validated US Lawful Permanent Resident persons access to CUI or export-controlled information. (Also, see DFARS 252.225-7048).

8.5 Reporting Requirements: Contractor personnel shall report to an appropriate authority any information or circumstances of which they are aware may pose a threat to the security of DoD personnel, contractor personnel, resources, and classified or unclassified defense information. Contractor employees shall be briefed by their immediate supervisor upon initial on-base assignment.

8.6 Physical Security: The contractor shall be responsible for safeguarding all Government property and controlled forms provided for contractor use. All Government facilities, equipment, and materials shall be secured when unoccupied or not in use.

8.7 Computer Security (COMPUSEC)/Automated Information System Requirements: If access to unclassified computer systems is required, the following requirements apply: Users must be a US citizen or receive AEDC or Air Force Materiel Command (AFMC) FDO approval prior to access being granted; requestor must have a favorable T1 investigation or an approved Interim Automated Information Systems request on file; and requestor must complete the AF Information Assurance Training. To request access to an unclassified computer system, users must complete a DD2875, System Authorization Access Request. Users bringing portable electronic devices on base and wanting to connect to AEDC computer systems/networks shall comply with Air Force Policy Directive (AFPD) 17-1, *Cyberspace*, AFI 17-130 *Cybersecurity Program Management*, and Chairman of the Joint Chiefs of Staff Instruction (CJCSI) 6510.01, *Information Assurance and Support to Computer Network Defense*. Employees shall complete any additional computer security training requirements mandated by the host base (e.g., Cyber Awareness Challenge).

8.8 Information Security: Information, regardless of media, shall be controlled and maintained IAW the National Industrial Security Program Operating Manual and guidance provided by the Servicing Security Activity, AEDC/TSD-IP.

8.9 Privacy Act: Comply with AFI 33-332, *Privacy Act Program*, when collecting and maintaining information protected by the Privacy Act of 1974, Title 5, US Code, Section 552a. Remove or destroy official records only IAW Air Force Records Information Management System, (AFRIMS) located on the internet at:
https://www.my.af.mil/gcssaf61a/afrims/afrims/rds/rds_series.cfm.

8.10 Operations Security (OPSEC): Comply with AFDPD 10-7, *Information Operations*; AFI 10-701, *Operations Security*; DoDM 5220.22, Volume 2, *National Industrial Security Program: Industrial Security Procedures for Government Activities*, Section 13; and local plans and publications.

8.11 Antiterrorism Training: DoD contractors shall be provided Antiterrorism information as required by DFARS Clause 252.225-7043, *Antiterrorism/Force Protection for Defense Contractors Outside the United States*. Level 1 Antiterrorism Awareness Training shall be provided to all DoD contractor employees.

8.12 Contractor Consent to Background Checks: The contractor shall not employ persons to perform under this contract if such employee is deemed or identified as a potential threat to the health, safety, security, general well-being, or operational mission of the installation and its population; nor shall the contractor or subcontractor employ persons under this contract who have an outstanding criminal warrant as determined through official Government records systems. Checks will be accomplished to verify if a person is wanted by local, state, or federal agencies. All contractor and subcontractor personnel must consent to background checks. Contractor and subcontractor personnel who do not consent to a check will be denied access to the installation. Information required to conduct a check includes full name, driver's license number, and/or social security number, date of birth of the person entering the installation, and completion of a background check questionnaire. The contractor must have this information ready to provide to the Visitor Control Center, if requested.

8.12.1 Contractors must ensure their employees and those of their subcontracts have the proper credentials allowing them to work in the US. Persons without proper credentials will be remanded to the appropriate authorities.

8.12.2 The contractor shall not be entitled to any compensation for delays or expenses associated with complying with the provisions of this requirement. Furthermore, nothing in this requirement shall excuse the contractor from proceeding with the contract as required.

8.13 Access to Installation During Force Protection Conditions (FPCONs): Gaining access to the installation during FPCONs will be done IAW the requirements outlined in the local Antiterrorism/Integrated Defense Plan(s). Access may be eliminated or strictly controlled as the FPCON increases.

8.14 Property Protection: Property protection for the facility where the contractor's primary work center is located will be the responsibility of the local facility manager and local security manager, or their duly authorized representative IAW AFI 31-101, *Integrated Defense*, and command/local directives to include AAFB Plan 31-101. The contractor must safeguard all Government-owned equipment and materials in their possession or use.

SECTION 9

9 ACRONYMS

AACEI	Association for the Advancement of Cost Engineering International
AAFB	Arnold Air Force Base
ADPS	Automated Data Processing Systems
AEDC	Arnold Engineering Development Complex
AEDCI	Arnold Engineering Development Complex Instruction
AEDCOI	Arnold Engineering Development Complex Operating Instruction
AF	Air Force
AF ETL	Air Force Engineering Technical Letter
AFB	Air Force Base
AFI	Air Force Instruction
AFMAN	Air Force Manual
AFPD	Air Force Policy Directive
AFTC	Air Force Test Center
ASME	American Society of Mechanical Engineers
BOE	Basis of Estimate
CADD	Computer-Aided Design and Drafting
CFR	Code of Federal Regulation
CO	Contracting Officer
COR	Contracting Officer Representative
CSI	Construction Specifications Institute
CUI	Controlled Unclassified Information
DAFFARS	Department of the Air Force Federal Acquisition Regulation Supplement
DAFGM	Department of the Air Force Guidance Memorandum
DAFI	Department of the Air Force Instruction
DAFMAN	Department of the Air Force Manual
DFARS	Defense Federal Acquisition Regulation Supplement
DLIS	Defense Logistics Information Service
DoD	Department of Defense

DoDM	Department of Defense Manual
EPA	Environmental Protection Agency
FAR	Federal Acquisition Regulation
FARM	Facilities Acquisitions for Restoration and Modernization
FDO	Foreign Disclosure Office
FEMP	Federal Energy Management Program
FFP	Firm-Fixed-Price
FOIA	Freedom of Information Act
FOPR	Fair Opportunity Proposal Request
FPCONs	Force Protection Conditions
FY	Fiscal Year
GSU	Geographically Separated Unit
HOOH	Home Office Overhead
IAW	In Accordance With
ID&C	Instrumentation, Data, & Control
IDIQ	Indefinite-Delivery Indefinite-Quantity
JOOH	Job Office Overhead
LPTA	Lowest Price Technically Acceptable
MIG	Mission Integration Group
MII	Micro-Computer Aided Cost Estimating System 2nd Generation
MIL-STD	Military Standard
MWP	Master Work Permit
NLT	No Later Than
NTP	Notice to Proceed
O&M	Operations and Maintenance
ODS	Ozone Depleting Substance
OSHA	Occupational Safety and Health Administration
P&ID	Piping and Instrumentation Diagram
PACES	Parametric Cost Engineering System Software
PDF	Portable Document Format
PE	Professional Engineering

PM	Program Manager
POC	Point of Contact
PoP	Period of Performance
QCP	Quality Control Plan
R/A	Reviewer/Approver
SDS	Safety Data Sheets
SFB	Space Force Base
SHE	Safety, Health, and Environment
SI	International System of Units
SOW	Statement of Work
SSPC	Society for Protective Coatings
STINFO	Scientific and Technical Information
TMDE	Test, Measurement, and Diagnostic Equipment
TO	Task Order
UFC	Unified Facilities Criteria
US	United States
WPS	Welding Procedure Specifications

SOW – ATTACHMENTS

**ATTACHMENT 1 – REFERENCES, REGULATORY, CODES AND
STANDARD DESIGN AND CONSTRUCTION CRITERIA**

ATTACHMENT 2 – DESIGN SUBMITTAL REQUIREMENTS

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ATTACHMENT 1 – REFERENCES, REGULATORY, CODES, AND STANDARD DESIGN AND CONSTRUCTION CRITERIA

The most applicable references are listed below, although others may apply. The contractor shall perform a code analysis to determine the specific codes applicable to each TO. In the event of a conflict between commercial and AF, regulations, codes, standards, instructions, technical letters, manuals, and handbooks, the more stringent criteria will apply. Proposed codes which meet the technical requirements will be considered, if in the best interest of the Government. Latest editions of the publications of the organizations listed below shall be used.

A1.1 FEDERAL, STATE, AND COMMERCIAL CODES AND STANDARDS

Acoustical Society of America (ASA)
Air-Conditioning and Refrigeration Institute (ARI)
American Association of State Highway and Transportation Officials (AASHTO)
American Concrete Institute (ACI)
American Industrial Hygiene Association (AIHA)
American Institute of Architects (AIA)
American Institute of Steel Construction (AISC)
American National Standards Institute (ANSI)
American Petroleum Institute (API)
American Society for Testing and Materials (ASTM)
American Society of Civil Engineers (ASCE)
American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE)
American Society of Mechanical Engineers (ASME)
American Society of Safety Engineers (ASSE)
American Water Works Association (AWWA)
American Welding Society (AWS)
Asphalt Institute (AI)
Builders Hardware Manufacturers Association (BHMA)
California Natural Resources Agency (CNRA)
Construction Specifications Institute (CSI)
Department of Labor - Code of Federal Regulation (CFR)
Electronic Industry Association (EIA)
Environmental Protection Agency (EPA)

Illuminating Engineering Society (IES)
Institute of Electrical and Electronics Engineers (IEEE)
Instrumentation Society Of America (ISA)
Interagency Committee on Seismic Safety in Construction (ICSSC)
International Building Code (IBC)
International Code Council (ICC)
InterNational Electrical Testing Association (NETA)
International Mechanical Code (IMC)
International Organization for Standardization (ISO)
International Plumbing Code (IPC)
Manufacturers Standardization Society of the Valve and Fittings Industry (MSS)
Maryland Department of Natural Resources (MDNR)
National Association of Corrosion Engineers (NACE)
National Association of Plumbing, Heating, and Cooling Contractors (NAPHCC)
National Concrete Masonry Association (NOIA)
National Earthquake Hazard Reduction Program (NEHRP)
National Electrical Code (NEC)
National Electrical Manufacturers Association (NEMA)
National Environmental Balancing Bureau (NEBB)
National Fire Protection Association (NFPA)
National Institute of Building Sciences (NIBS)
National Institute of Standards and Technology (NIST)
National Sanitation Foundation (NSF)
Occupational Safety and Health Administration (OSHA)
Sheet Metal and Air-Conditioning Contractors National Association (SMACNA)
Society for Protective Coatings (SSPC)
South Coast Air Quality Management District (SCAQMD)
Steel Structures Painting Council (SSPC)
Telecommunications Industry Association (TIA)
Tennessee Department of Environment and Conservation (TDEC)
Underwriter's Laboratories (UL)

Unified Facilities Criteria (UFC)

A1.2 AEDC OPERATING INSTRUCTIONS (AEDC OIs) AND STANDARDS

AEDC-ENGR-STD-T-1, AEDC Standard Pressure Vessels
AEDC-ENGR-STD-T-2, AEDC Standard Pressure Piping
AEDC-ENGR-STD-T-3, AEDC Standard Engineering Drawing and Drafting Practices
AEDC-ENGR-STD-T-4, AEDC Standard For Procurement Documentation
AEDC-ENGR-STD-T-5, AEDC Standard Welding Practices
AEDCI 21-3, Tool Control
AEDCI 63-3733, Organizational Engineering
AEDCOI 21-1, Foreign Object Damage Prevention Program
AEDCOI 21-2, Hold and Impoundment
AEDCOI 21-205, Tactical Integration Group
AEDC-STD-CM-1, Configuration Management
AFTC Instruction 91-203 AEDC Supplement, AFTC Test Safety Review Policy
Arnold AFB Architectural Compatibility Standards
Asbestos Removal CSI Section 02 08 00 modified for AEDC
Corrosion Control Plan (CCP)-100, AEDC Protective Coatings and Corrosion Control Plan
Lead and Heavy Metal Removal CSI Section 02 08 50 modified for AEDC
TSS-SE-HNDBK, Test Systems Sustainment Systems Engineering Handbook

A1.3 AEDC SHE STANDARDS

A4, System Safety
A6, User and Subcontractor Safety
A9, Hazard Communication
A10 Job Safety Analysis
B1, Master Work Permit
B2, Lockout/Tagout
B3, Control of Hazardous Areas Using Safety Signs, Tags, and Barricades
B5, Confined Spaces
B6, Low Voltage Electrical Safety Related Work Practices
D3, Identification of Piping Systems

D6, Aerial Work Platforms
D8, Portable Power Tools
D9, Fixed and Portable Ladders
D10, Scaffolds
E7, Asbestos
E9, Wastewater Discharge Management
E17, Oil and Hazardous Substances Spill Response
E18, Managing Wastes Containing Chemical or Petroleum Products
E19, Lead and Heavy Metals
F2, Personal Protection Equipment
F6, Fall Protection

A1.4 Regulations

10 CFR 435 Energy Conservation Voluntary Performance Standards for New Buildings.
Mandatory for Federal Buildings
DAFFARS 5352.223-9000, Elimination of Use of Class I Ozone Depleting Substances
(ODS), October 2024
FAR 52.223-9, Estimate of Percentage of Recovered Material Content for EPA-
Designated Items, October 2025
Energy Policy Act

A1.5 AIR FORCE ENGINEERING TECHNICAL LETTERS (AF ETLs)

AF ETL 01-1, Reliability and Maintainability Design Checklist
AF ETL 01-18, Fire Protection Engineering Criteria-Electronic Equipment Installations,
24 October 2001
AF ETL 02-12, Communications and Information Systems Criteria for Air Force
Facilities, 2008
AF ETL 08-13, Incorporating Sustainable Design and Development (SDD) and Facility
Energy Attributes in the Air Force Construction Program, 14 September 2008
AF ETL 11-21, Emergency and Standby Generator Design, Maintenance and Testing
Criteria, 2011

A1.6 AFIs, DEPARTMENT OF THE AIR FORCE INSTRUCTIONS (DAFIs), AFMANs, DEPARTMENT OF THE AIR FORCE MANUALS (DAFMANs), DEPARTMENT OF THE AIR FORCE GUIDANCE MEMORANDUMS (DAFGMs), AND MILITARY STANDARDS (MIL-STDs):

AFMAN 32-1068, Heating Systems and Unfired Pressure Vessels

AFMAN 91-201 Explosives Safety Standards

DAFGM2024-32-01, Civil Engineer Control Systems Cybersecurity

DAFI 32-2001, Fire and Emergency Services Program

DAFI 32-1020, Planning and Programming Built Infrastructure Projects

DAFMAN 32-1084, Standard Facility Requirements

DAFMAN 32-7002, Environmental Compliance and Pollution Prevention

MIL-STD-3007, Unified Facilities Criteria, Facilities Criteria and Unified Facilities
Guide Specifications

END OF ATTACHMENT 1

ATTACHMENT 2 – DESIGN SUBMITTAL REQUIREMENTS

A2.1 30-PERCENT DESIGN

A2.1.1 The 30-percent design submission shall include, but not be limited to, the following items:

- a) Concept design narrative
- b) Concept drawings
- c) Project specifications outline
- d) Risk assessment
- e) Preliminary hazards list
- f) Preliminary schedule
- g) List of applicable codes and standards based on code analysis

A2.1.2 Concept design narrative:

- a) Design approach that addresses major disciplines associated with the design solution
- b) System and subsystem requirements, including derived requirements, traceable back to TO documentation
- c) Assumptions and constraints
- d) O&M impacts
- e) Analysis and calculations

A2.1.3 Concept drawings, as applicable:

- a) Mechanical/Electrical/ID&C:
 - 1) Location plan and general characteristics of major equipment
 - 2) Preliminary layout of ducts/piping/conduits
 - 3) Preliminary Piping and Instrumentation Diagram (P&ID)
- b) Civil:
 - 1) Site plan showing existing drainage, roads and sidewalks, parking and routing of electric, telephone, gas, water, sewer, and other utilities
 - 2) Footings and foundations, including structural
 - 3) Geotechnical
- c) Demolition plans generally depicting equipment, utilities and site infrastructure to be removed with interface points

A2.1.4 The risk assessment shall identify high-level risks and mitigation measures.

A2.1.5 A list shall be provided of preliminary safety hazards resulting from changes introduced by the design. (Reference SHE standard A4 System Safety, Annex H - Hazards Checklist)

A2.1.6 The project specifications shall be in CSI format and outline all applicable sections. Project specifications shall outline and be inclusive of all applicable disciplines, codes and standards references in the design concept narrative. Specifications selected due to an identified risk, requirement, assumption, and/or constraint shall be noted as such.

A2.1.7 A preliminary schedule of activities shall be provided with emphasis on how the work will be broken down and progress measured. A Gantt chart shall be provided that depicts the logical sequence of work from design through closeout including major activities, milestones, outages, start and completion dates and predecessors and successors to depict the critical path.

A2.1.8 All applicable engineering codes and standards shall be provided as they apply to each system.

A2.2 60-PERCENT DESIGN

A2.2.1 The preceding design shall be completed to 60 percent and shall include the integration of the Government review comments resulting from the previous review. The 60-percent design submission shall include, but not be limited to, the following items:

- a) Design drawings with appropriate detail
- b) Design documentation/data
- c) Project specifications
- d) Commissioning Plan outline
- e) Risk Management Plan outline
- f) Preliminary hazards analysis
- g) List of training requirements
- h) Schedule
- i) List of applicable codes and standards based on code analysis

A2.2.2 Design Drawings:

- a) Preliminary site plans shall show the location of each proposed building and facility, critical grades, contours, drainage, and relation of buildings and facilities to existing buildings, facilities, street systems, above and underground utilities, entrances, and finished grades. Present grades shall be indicated. At locations where survey controls (coordinate systems or base lines) have been established, buildings and facilities shall be tied-in to such control. Logs of subsurface explorations shall be shown. (As applicable)
- b) Preliminary designs for buildings and structures/equipment shall be two-line drawings showing floor plans, elevations, and locations of principal items of installed equipment IAW the following, as applicable:

- 1) All required controlling dimensions shall be clearly indicated.

2) Floor plans with dimensions and functional arrangement of all areas including corridors, exits, stairs and utility spaces properly related to exterior access, road, parking, and service areas. Individual treatment shall be given to items involving special design and/or deviations from accepted standards and of complex design. Gross floor areas shall be shown for each floor and for the entire building. Floor and roof framing and loadings shall be indicated to scale. Pertinent information regarding fire prevention and safety requirements shall be shown. Design, dead, live, snow, wind, seismic, any special loading, and all load combinations shall be shown.

3) Elevations shall include story heights, fenestration, and adaptation to finished grades at the site.

4) Sections of structures, including framing, partitions, suspended ceilings, and ducts shall be shown.

5) Typical wall sections shall be shown at a scale of no less than $\frac{3}{4}$ inch = 1 foot or engineering scale equivalent. Sections shall include material and thicknesses, methods of attachment, and type of windows with relation thereof to supporting structural columns or walls.

c) Preliminary designs for mechanical systems to include, but are not limited to: high-pressure air ducting, valves and associated equipment; plant mechanical equipment; cooling water lines, valves and associated equipment; refrigeration; fire protection; hoists and cranes; and special mechanical features.

1) Location, capacity and space requirements of all major items of equipment shall be indicated.

2) Whether piping and ducting is exposed or concealed is required. Functional areas where piping is exposed shall also be delineated.

3) Installation floor plans shall show piping as single line with pipe size and service delineated adjacent to pipe. Piping in areas with physical constraints shall be shown as double line and shall include critical dimensions.

4) Elevations shall be provided of major equipment items showing connecting ductwork and piping. Elevations shall be shown as required to clarify physical constraints and delineate critical dimensions.

5) Details shall be provided for equipment mounting, equipment bases, and equipment supports.

6) Details shall be provided for floor, roof, and wall penetrations.

7) Welding symbols shall be included when welding is required.

8) Piping schematics shall be provided for each system.

9) Schedules shall be provided for equipment including but not limited to chillers, fans, pumps, control valves, piping specialties, and variable frequency drives. Schedules shall include device or equipment number and pertinent information

such as type, size, flow coefficient, flow rate, pattern, design conditions, operating parameters, performance requirements, and suggested manufacturer and model numbers.

10) Drawings shall include a legend that clearly identifies each symbol used on the mechanical drawings.

11) Drawings shall also include all notes required for installation.

d) Preliminary designs for electrical systems to include, but are not limited to: electrical service, distribution, and all related equipment.

1) Location, physical capacity, and space requirements of all major items of equipment shall be shown including interior unit substations, motor control breakers/centers, service equipment, and panels.

2) Electrical load analysis shall be provided,

3) Elevations shall be provided of major equipment items showing connecting conduit and wiring. Elevations shall be shown as required to clarify physical constraints and delineate critical dimensions.

4) Details shall be provided for equipment mounting, equipment bases, and equipment supports.

5) Details shall be provided for floor, roof, and wall penetrations.

6) One-line diagrams of high-voltage and low-voltage systems shall indicate equipment-connect load, circuit breaker and disconnect switch ratings, circuit breaker trip settings, fuse sizes and types, and wire sizes to each load.

7) Detailed schematics and wiring diagrams shall show interconnecting wires and cabling between individual components and equipment, and wiring up to the AEDC interfaces. Details as a minimum shall include identification of all contractor-provided wire and cable as to type, number of conductors, shielding and color code (where applicable). Each cable and individual wire shall have a unique identification. Individual terminal points on terminal blocks shall be numbered.

8) Conduits, including those to be wired by others, shall be shown together with indication as to whether conduits are exposed. Functional areas having exposed conduits shall be delineated.

9) Riser diagrams showing service equipment, feeders and panels, other than branch circuits shall be shown. Power, communication and electronic features shall be indicated in sufficient detail to identify requirements clearly and to indicate the means of satisfying the requirements.

10) Schedules shall be provided for equipment. Schedules shall include device or equipment number and pertinent information such as type, size, design conditions, operating parameters, performance requirements, and suggested manufacturer and model numbers.

11) Drawings shall include a legend that clearly identifies each symbol used on the mechanical drawings.

12) Drawings shall also include all notes required for installation.

e) P&ID control schematics for all equipment shall be provided showing all end devices. A sequence of operations shall be provided on the drawings for each major equipment item. The sequence of operation shall provide detailed explanations for all control logic and shall be written in present tense.

f) Printouts of Graphic Displays shall be provided along with all software code and related items for any user interface displays.

g) Required information assurance documentation shall be prepared, to include network diagrams and system security configuration documentation, as specified within the TO. This documentation will support the submission of required accreditation.

h) Demolition plans and elevations shall clearly delineate the equipment, utilities and site infrastructure to be removed with clear interface points.

i) Note any Government Furnished Equipment/Government Furnished Property to be installed by the contractor, and equipment to be furnished and installed by the Government.

A2.2.3 Design documentation/data shall:

a) Document requirements traceability from TO requirements to written specifications

b) Document O&M planning

c) Address mechanical/electrical design criteria and conditions being utilized during analysis

d) Address civil/structural areas to include, but not limited to:

1) Geotechnical Reports, boring plans and logs, including allowable bearing capacities, and friction bearing capacities;

2) Footing and foundation structural calculations;

3) Design loads (dead, live, snow, wind, seismic, any special loading and all load combinations); and

4) All structural calculations.

e) Document routing and capacities of existing/proposed cooling water, power, air ducting, communication and fire alarm systems; loads, capacities and general arrangement, and illumination level of proposed lighting

A2.2.4 Detailed specifications shall cover all proposed construction materials and equipment to be incorporated in the project as the result of designs by the contractor. Detailed specifications shall be coordinated with the requirements for detailed design drawings. Any proposed deviations from the standard guide specifications shall be described.

A2.2.5 The Commissioning Plan outline shall identify all major equipment and systems requiring Verification, Validation, and Accreditation.

A2.2.6 Risk Management Plan outline shall:

- a) Identify cost, schedule and performance risks, beyond the high-level risks previously identified
- b) Thoroughly describe mitigation strategies for each significant risk

A2.2.7 An analysis of the preliminary safety hazards resulting from changes introduced by the design shall be provided. The contractor shall elaborate on proposed mitigation measures for the preliminary hazards listed in the 30-percent design and those resulting from the 60-percent design effort.

A2.2.8 A list of the training requirements shall be provided to include training materials and sessions necessary to train O&M personnel on any system modifications that impact O&M practices.

A2.2.9 An updated schedule shall be provided. Activities shall be divided into measureable tasks useful for progress assessments. Further breakdown may be required to include interactions between the contractor and Government, such as submittal approvals ahead of associated work, design reviews, large material deliveries, on-site work, inspections, and system outages. The contractor shall ensure relationships (predecessors and successors) between major activities, milestones, Government/contractor interaction, start and completion dates accurately represent how the work will be performed.

A2.2.10 A requirements traceability matrix shall be provided to show that all TO requirements have been fulfilled.

A2.2.11 An updated code analysis shall be provided.

A2.3 90-PERCENT DESIGN

A2.3.1 The preceding design shall be completed to 90 percent and shall include the integration of the Government review comments resulting from the previous reviews. The 90-percent design submission shall include, but not be limited to, the following items:

- a) Completed design drawings
- b) Completed design documentation/data
- c) Completed project specifications
- d) Commissioning Plan
- e) Risk Management Plan
- f) Training Plan
- g) Updated schedule
- h) Finalized list of applicable codes and standards

A2.3.2 The Commissioning Plan shall be updated to address the steps and methods to be used, as well as required AEDC operational support. Additionally, the commissioning plan shall include estimated timing/timeline for each major step of the plan.

A2.3.3 The Risk Management Plan shall be updated for cost, schedule and performance risks and mitigation strategies for each significant risk.

A2.4 FINAL DESIGN

The final design shall include all items in the 90-percent Design Submittal and any necessary revisions to provide a complete, comprehensive submittal. Drawings shall be 100 percent complete, stamped, and signed by the appropriate registered discipline that performed the design. Registration shall be in the state in which the work is located.

END OF ATTACHMENT 2